CIE A-Level Physics

Revision Booklet

This booklet covers:

- Topic 13 — Gravitational Fields
- Topics 14, 15 & 16 — Thermal Physics
- Topic 17 — Oscillations
- Topic 18 — Electric Fields
- Topic 19 — Capacitance
- Topic 20 — Magnetic Fields
- Topic 21 — Alternating Currents
- Topic 22 — Quantum Physics
- Topic 23 — Nuclear Physics
- Topic 24 — Medical Physics
- Topic 25 — Astronomy and Cosmology

Contents

Gravitational Fields **3**

Thermal Physics **18**

Oscillations **33**

Electric Fields **47**

Capacitance **61**

Magnetic Fields **74**

Alternating Currents **86**

Quantum Physics **97**

Nuclear Physics **100**

Medical Physics **103**

Astronomy and Cosmology **106**

CIE International A Level Physics

Topic 13

Gravitational Fields

Revision Booklet

This booklet covers:

- Newton's Law of Gravitation
- Gravitational Field Strength
- Gravitational Potential
- Orbital Motion & Kepler's Third Law
- Escape Velocity

1	Core Concepts and Definitions	5
2	Newton's Law of Gravitation	5
3	Gravitational Field Strength	6
4	Gravitational Potential	7
5	Circular Orbits and Satellites	8
6	Escape Velocity	9
7	Formula Summary Sheet	10
8	Exam Technique and Problem-Solving Strategy	10
9	Worked Examples	11
10	Practice Exam Questions	12
11	Mark Scheme and Answers	15
12	Revision Checklist	17

1 Core Concepts and Definitions

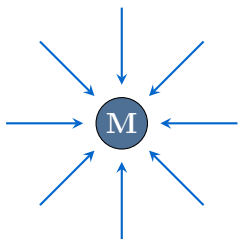
Gravitational Field

A **gravitational field** is a region of space in which a mass experiences a force due to the gravitational attraction of another mass.

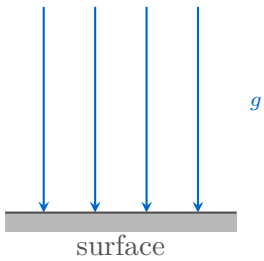
- Gravitational fields are always **attractive** — there is no gravitational repulsion.
- Any mass creates a gravitational field in the space around it.
- A **test mass** placed in the field will experience a force towards the source mass.

Radial vs Uniform Fields

- **Radial field** (e.g., around a planet): field lines point towards the centre; g decreases with distance.
- **Uniform field** (e.g., near Earth's surface over small distances): field lines are parallel and equally spaced; g is approximately constant at 9.81 N kg^{-1}.



Radial Field



Uniform Field

2 Newton's Law of Gravitation

Newton's Law of Gravitation

$$F = \frac{Gm_1m_2}{r^2}$$

F	= gravitational force between the two masses (N)
G	= gravitational constant = 6.67×10^{-11} N m^{2}kg^{-2}
m_1, m_2	= the two point masses (or uniform spheres) (kg)
r	= separation between centres of mass (m)

Key Points

- The force is always **attractive**.
- It applies strictly to **point masses**, but also works for uniform spheres if r is measured from the centre.
- r is the **centre-to-centre** distance — not surface to surface.

- The force obeys an **inverse-square law**: double the distance, quarter the force.

3 Gravitational Field Strength

Definition of Gravitational Field Strength g

The **gravitational field strength** at a point is the gravitational force exerted per unit mass on a small test mass placed at that point.

$$g = \frac{F}{m} \qquad \text{units: N kg}^{-1} \equiv \text{m s}^{-2}$$

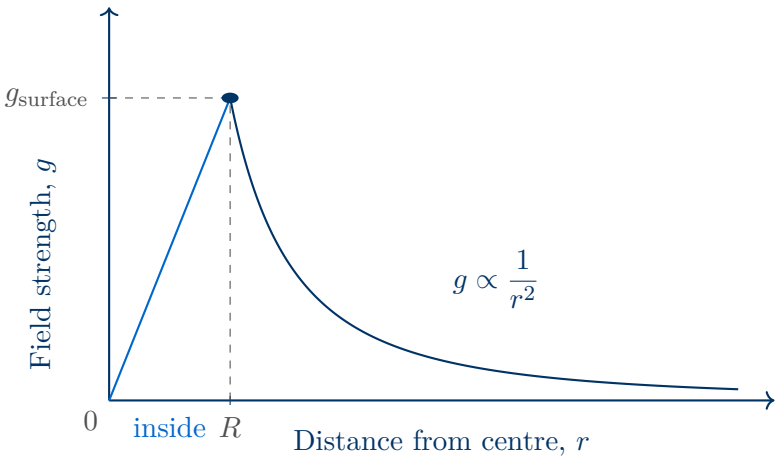
g is a **vector** quantity, directed towards the source mass.

Field Strength Formulae

$$g = \frac{GM}{r^2}$$

g = gravitational field strength at distance r (N kg^{-1})
G = gravitational constant (N m^2 kg^{-2})
M = mass of body creating the field (kg)
r = distance from centre of M (m)

Variation of g with distance r



Common Mistake

Do not confuse g (field strength, N kg^{-1}) with G (the universal gravitational constant, N m^2 kg^{-2}). They are completely different quantities!

4 Gravitational Potential

Definition of Gravitational Potential ϕ

The **gravitational potential** at a point is the work done per unit mass to move a small test mass from infinity to that point.

$$\phi = \frac{W}{m} \qquad \text{units: J kg}^{-1}$$

Gravitational potential is always **negative** (zero at infinity; work is done *by* gravity as mass moves inwards, so the system loses potential energy).

Gravitational Potential Formulae

$$\phi = -\frac{GM}{r} \qquad E_p = m\phi = -\frac{GMm}{r}$$

ϕ = gravitational potential at distance r (J kg^{-1})
E_p = gravitational potential energy (J)
G = 6.67×10^{-11} N m^2 kg^{-2}
M = mass of body creating the field (kg)
r = distance from centre of M (m)

4.1 Relationship Between g and ϕ

Field Strength from Potential Gradient

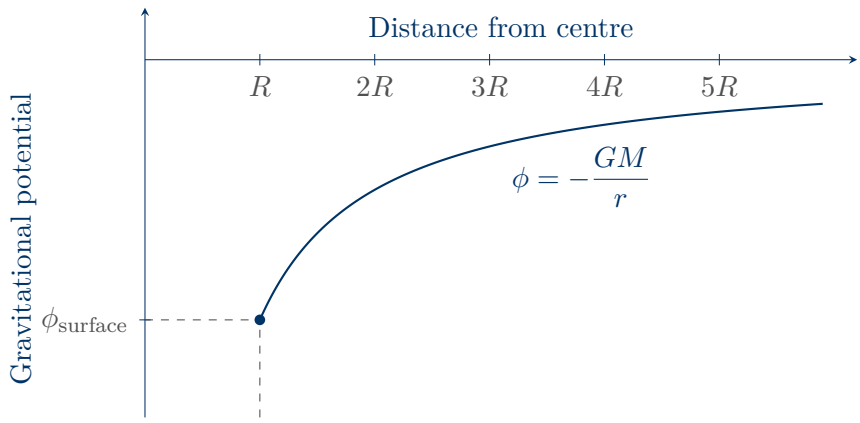
$$g = -\frac{\Delta\phi}{\Delta r}$$

Interpreting the Gradient

g is the **negative gradient** of the potential–distance graph. On a ϕ–r graph:

- A steeper gradient ⇒ stronger field (larger g)
- The gradient becomes shallower as r increases ⇒ g decreases
- The area under a g–r graph gives the change in potential $\Delta\phi$

Graph of ϕ against r



$\phi \to 0$ as $r \to \infty$

Equipotential Surfaces

An **equipotential surface** is a surface on which the gravitational potential is the same everywhere.

- No work is done moving a mass along an equipotential surface.
- Equipotentials are always **perpendicular** to field lines.
- For a point mass (or uniform sphere), equipotentials are concentric spheres.

5 Circular Orbits and Satellites

For an object of mass m in a circular orbit of radius r around a mass M, the gravitational force provides the centripetal force:

Orbital Mechanics

$$\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2$$

$$\therefore \quad v = \sqrt{\frac{GM}{r}} \qquad \text{(orbital speed)}$$

$$T^2 = \frac{4\pi^2}{GM}r^3 \qquad \text{(Kepler's Third Law)}$$

Kepler's Third Law

$$T^2 \propto r^3$$

The square of the orbital period is proportional to the cube of the orbital radius. The constant of proportionality is $\frac{4\pi^2}{GM}$.

5.1 Energy in Circular Orbits

Energy of a Satellite in Circular Orbit

$$E_k = \frac{1}{2}mv^2 = \frac{GMm}{2r} \qquad \text{(kinetic energy — always positive)}$$
$$E_p = -\frac{GMm}{r} \qquad \text{(potential energy — always negative)}$$
$$E_{total} = E_k + E_p = -\frac{GMm}{2r} \qquad \text{(total energy — always negative)}$$

As r increases: v decreases, E_k decreases, E_p increases, E_{total} increases (becomes less negative).

5.2 Geostationary Orbits

Geostationary Satellite

A **geostationary** satellite has:

- An orbital period of exactly **24 hours**
- An orbit in the **equatorial plane**
- Movement in the **same direction** as Earth's rotation (west to east)
- An orbital radius of approximately 4.2×10^7 m ($\approx 36\,000$ km altitude)

It appears **stationary** relative to a point on Earth's surface.

Uses: satellite TV, telecommunications, weather monitoring.

6 Escape Velocity

Definition of Escape Velocity

The **escape velocity** is the minimum speed at which an object must be projected from the surface of a body so that it can escape to infinity against the gravitational field, without any further energy input.

Escape Velocity

Setting total energy = 0 at infinity:

$$\frac{1}{2}mv_{esc}^2 - \frac{GMm}{R} = 0$$

$$v_{esc} = \sqrt{\frac{2GM}{R}}$$

Note: $v_{esc} = \sqrt{2}\,v_{orbital}$ at the same radius.

7 Formula Summary Sheet

Formula	Quantity	Units
$F = \dfrac{Gm_1m_2}{r^2}$	Gravitational force	N
$g = \dfrac{F}{m}$	Field strength (definition)	N kg^{-1}
$g = \dfrac{GM}{r^2}$	Field strength (point mass)	N kg^{-1}
$\phi = -\dfrac{GM}{r}$	Gravitational potential	J kg^{-1}
$E_p = -\dfrac{GMm}{r}$	Gravitational PE	J
$g = -\dfrac{\Delta\phi}{\Delta r}$	Field from potential gradient	N kg^{-1}
$v = \sqrt{\dfrac{GM}{r}}$	Orbital speed	m s^{-1}
$T^2 = \dfrac{4\pi^2r^3}{GM}$	Kepler's Third Law	s^2, m^3
$v_{esc} = \sqrt{\dfrac{2GM}{R}}$	Escape velocity	m s^{-1}

Constants: $G = 6.67 \times 10^{-11}$ N m^{2}kg^{-2}, $M_E = 5.97 \times 10^{24}$ kg, $R_E = 6.37 \times 10^6$ m

8 Exam Technique and Problem-Solving Strategy

Step-by-Step Strategy for Calculation Questions

1. **Identify** what you are asked to find.
2. **List** the quantities given; convert units if needed (e.g. days → seconds, km → m).
3. **Select** the appropriate formula.
4. **Substitute** values carefully, showing all working.
5. **Check** units and significant figures in your answer.

Common Errors — Avoid These!

- Using **diameter** instead of radius in $g = GM/r^2$.
- Forgetting the **negative sign** in $\phi = -GM/r$.
- Not converting **hours/days to seconds** before applying Kepler's Law.
- Confusing g (field strength, vector) with ϕ (potential, scalar).
- Saying $\phi = 0$ at the surface — it is zero only **at infinity**.

- Thinking higher orbit ⇒ faster orbital speed — it is **slower**.

9 Worked Examples

Example 1 — Field Strength at Altitude

Question: Calculate the gravitational field strength at a height of 400 km above Earth's surface. ($M_E = 5.97 \times 10^{24}$ kg, $R_E = 6.37 \times 10^6$ m)

Solution

Solution:
$r = R_E + h = 6.37 \times 10^6 + 4.00 \times 10^5 = 6.77 \times 10^6$ m

$$g = \frac{GM}{r^2} = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{(6.77 \times 10^6)^2}$$

$$g = \frac{3.98 \times 10^{14}}{4.58 \times 10^{13}} = \mathbf{8.69}\ \text{N kg}^{-1}$$

Note: this is the field strength at the ISS orbit — astronauts are *not* weightless because gravity is zero!

Example 2 — Orbital Period Using Kepler's Third Law

Question: A satellite orbits Earth at a radius of 4.2×10^7 m. Calculate its orbital period.

Solution

Solution:

$$T^2 = \frac{4\pi^2 r^3}{GM} = \frac{4\pi^2 \times (4.2 \times 10^7)^3}{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}$$

$$T^2 = \frac{4\pi^2 \times 7.41 \times 10^{22}}{3.98 \times 10^{14}} = 7.36 \times 10^9\ \text{s}^2$$

$$T = \sqrt{7.36 \times 10^9} = 8.58 \times 10^4\ \text{s} \approx \mathbf{23.8}\ \text{hours}$$

This is a geostationary orbit ($T \approx 24$ h).

Example 3 — Change in Gravitational Potential Energy

Question: A spacecraft of mass 2500 kg is moved from the Earth's surface to an altitude of 800 km. Calculate the increase in gravitational potential energy.

Solution

Solution:
$r_1 = 6.37 \times 10^6$ m, $\quad r_2 = 6.37 \times 10^6 + 8.00 \times 10^5 = 7.17 \times 10^6$ m

$$\Delta E_p = GMm\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$\Delta E_p = 6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times 2500 \times \left(\frac{1}{6.37 \times 10^6} - \frac{1}{7.17 \times 10^6}\right)$$

$$\Delta E_p = 9.95 \times 10^{17} \times 1.75 \times 10^{-8} = \mathbf{1.74 \times 10^{10}}\ \text{J}$$

10 Practice Exam Questions

Section A — Short Answer Questions

Q1. State Newton's Law of Gravitation.

[2 marks]

Q2. Define gravitational field strength and state its units.

[2 marks]

Q3. Explain why gravitational potential is always a negative quantity.

[2 marks]

Q4. The gravitational field strength at Earth's surface is 9.81 N kg^{-1}. Calculate the field strength at a distance of $3R_E$ from Earth's centre, where R_E is Earth's radius.

[2 marks]

Q5. Two planets, X and Y, orbit the same star. Planet X has an orbital radius twice that of planet Y. Determine the ratio T_X/T_Y.

[3 marks]

Section B — Longer Structured Questions

Q6. A satellite of mass m orbits a planet of mass M in a circular orbit of radius r.

(a) Show that the orbital speed of the satellite is given by $v = \sqrt{GM/r}$.

[2 marks]

(b) Show that the period of the orbit satisfies $T^2 = \dfrac{4\pi^2 r^3}{GM}$.

[2 marks]

(c) The satellite is moved to a lower orbit. Explain what happens to its:

- speed
- kinetic energy
- total energy

[3 marks]

Q7. The Moon orbits the Earth with a period of 27.3 days at a mean orbital radius of 3.84×10^8 m.

(a) Use this data to calculate the mass of the Earth.

[3 marks]

(b) Calculate the gravitational potential at the Moon's orbital radius.

[2 marks]

(c) A 75 kg astronaut travels from Earth's surface to the Moon's orbital radius. Calculate the change in gravitational potential energy.

[3 marks]

11 Mark Scheme and Answers

Q1. Any two masses exert an attractive force on each other [1]; the force is proportional to the product of their masses and inversely proportional to the square of their separation [1].

Q2. Gravitational field strength is the gravitational force per unit mass acting on a (small test) mass placed at that point [1]; units: N kg^{-1} (or m s^{-2}) [1].

Q3. Gravitational potential is defined as zero at infinity [1]; work is done by gravity as mass moves from infinity inward, so the potential decreases below zero — it is always negative [1].

Q4. $g \propto 1/r^2$; at $3R_E$, $g = 9.81/3^2 = 9.81/9 = \mathbf{1.09}$ N kg$^{\mathbf{-1}}$ [2].

Q5. By Kepler's Third Law: $T^2 \propto r^3$, so $\left(\frac{T_X}{T_Y}\right)^2 = \left(\frac{2r_Y}{r_Y}\right)^3 = 8$ [2]; $\frac{T_X}{T_Y} = \sqrt{8} = \mathbf{2\sqrt{2} \approx 2.83}$ [1].

Q6(a). Gravitational force = centripetal force: $\frac{GMm}{r^2} = \frac{mv^2}{r}$ [1]; cancel m, rearrange: $v = \sqrt{GM/r}$ [1].

Q6(b). Substitute $v = 2\pi r/T$ into result from (a) [1]; rearrange to get $T^2 = 4\pi^2 r^3/GM$ [1].

Q6(c). Speed **increases** [1]; KE **increases** [1]; total energy **decreases** (becomes more negative) [1].

Q7(a). $T = 27.3 \times 24 \times 3600 = 2.36 \times 10^6$ s; $M = \frac{4\pi^2 r^3}{GT^2} = \frac{4\pi^2(3.84 \times 10^8)^3}{6.67 \times 10^{-11}(2.36 \times 10^6)^2}$ [1] $= \mathbf{6.02 \times 10^{24}}$ kg [2].

Q7(b). $\phi = -\frac{GM}{r} = -\frac{6.67 \times 10^{-11} \times 6.02 \times 10^{24}}{3.84 \times 10^8} = \mathbf{-1.05 \times 10^6}$ J kg$^{\mathbf{-1}}$ [2].

Q7(c). $\Delta E_p = m\Delta\phi = m(\phi_{\text{Moon orbit}} - \phi_{\text{surface}})$ [1]; $\phi_{\text{surface}} = -GM/R_E = -6.25 \times 10^7$ J kg^{-1}; $\Delta\phi = -1.05\times10^6 - (-6.25\times10^7) = 6.14\times10^7$ J kg^{-1}; $\Delta E_p = 75\times6.14\times10^7 = \mathbf{4.6 \times 10^9}$ J [2].

12 Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

	Learning Objective	Confidence (1–3)
☐	State and apply Newton's Law of Gravitation	
☐	Define gravitational field strength; use $g = F/m$ and $g = GM/r^2$	
☐	Sketch field line and equipotential diagrams	
☐	Define gravitational potential and explain why it is negative	
☐	Use $\phi = -GM/r$ and $E_p = -GMm/r$	
☐	Apply the relationship $g = -\Delta\phi/\Delta r$	
☐	Derive the expression for orbital speed	
☐	State and apply Kepler's Third Law ($T^2 \propto r^3$)	
☐	Analyse energy changes in satellite orbits	
☐	Describe the properties of a geostationary satellite	
☐	Derive and apply the formula for escape velocity	
☐	Interpret graphs of g vs r and ϕ vs r	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

Remember: understanding *why* formulas work is more powerful than memorising them. Practice drawing diagrams and deriving key results from first principles.

CIE International A Level Physics

Topic 14, 15 & 16

Thermal Physics

Revision Booklet

This booklet covers:

- Temperature and Thermal Equilibrium
- Temperature Scales
- Specific Heat Capacity and Latent Heat
- The Mole and Ideal Gases
- Kinetic Theory of Gases
- Internal Energy
- The First Law of Thermodynamics

13	Thermal Equilibrium	20
14	Temperature Scales	20
15	Specific Heat Capacity and Latent Heat	21
16	The Mole and Ideal Gases	22
17	Kinetic Theory of Gases	23
18	Internal Energy	25
19	The First Law of Thermodynamics	25
20	Formula Summary	27
21	Worked Examples	27
22	Practice Exam Questions	28
23	Mark Scheme and Answers	31
24	Revision Checklist	32

13 Thermal Equilibrium

Thermal Energy Transfer

Thermal (heat) energy is transferred from a region of **higher temperature** to a region of **lower temperature**. Transfer continues until the two regions reach the same temperature.

Thermal Equilibrium

Two objects are in **thermal equilibrium** when there is no net transfer of thermal energy between them. This occurs when they are at the **same temperature**.

Zeroth Law of Thermodynamics: If object A is in thermal equilibrium with object C, and object B is also in thermal equilibrium with object C, then A and B are in thermal equilibrium with each other.

14 Temperature Scales

Thermometric Property

A **thermometric property** is a physical property that varies continuously and measurably with temperature. Examples include:

- Resistance of a metal wire (e.g. platinum resistance thermometer)
- e.m.f. of a thermocouple
- Volume of a gas at constant pressure
- Density of a liquid

Thermodynamic Temperature Scale

The **thermodynamic (Kelvin) temperature scale** does not depend on the property of any particular substance. It is an absolute scale with:

- **Absolute zero** (0 K) — the lowest possible temperature
- **Triple point of water** (273.16 K) — fixed reference point

Temperature Conversion

$$T/\mathrm{K} = \theta/^\circ\mathrm{C} + 273.15$$

T = thermodynamic temperature (K)
θ = Celsius temperature (°C)

Note: a temperature *difference* of 1 K equals a difference of 1 °C.

Absolute Zero

At absolute zero (0 K = −273.15 °C):

- Molecules have minimum possible kinetic energy
- It is impossible to reach in practice, only to approach asymptotically
- All ideal gases would have zero volume (zero pressure at constant volume)

15 Specific Heat Capacity and Latent Heat

Specific Heat Capacity

The **specific heat capacity** c of a substance is the energy required to raise the temperature of **unit mass** by **one kelvin** (or one degree Celsius), without change of state.

$$c = \frac{Q}{m\,\Delta T} \qquad \text{units: } \mathrm{J\,kg^{-1}K^{-1}}$$

Specific Heat Capacity

$$Q = mc\Delta T$$

Q = thermal energy transferred (J)
m = mass (kg)
c = specific heat capacity ($\mathrm{J\,kg^{-1}\,K^{-1}}$)
ΔT = temperature change (K or °C)

Specific Latent Heat

The **specific latent heat** L of a substance is the energy required to change the state of **unit mass** at constant temperature.

$$L = \frac{Q}{m} \qquad \text{units: } \mathrm{J\,kg^{-1}}$$

- **Specific latent heat of fusion** L_f: solid → liquid (melting)
- **Specific latent heat of vaporisation** L_v: liquid → gas (boiling)

Note: $L_v > L_f$ for any given substance, since greater work is done separating molecules completely during vaporisation.

Specific Latent Heat

$$Q = mL$$

Q = thermal energy transferred (J)
m = mass (kg)
L = specific latent heat ($\mathrm{J\,kg^{-1}}$)

Common Mistake

During a change of state, temperature remains **constant** even though energy is being supplied. The energy goes into breaking intermolecular bonds, not increasing kinetic energy.

16 The Mole and Ideal Gases

Amount of Substance

The **mole** (mol) is the SI base unit for amount of substance. One mole of any substance contains exactly $N_A = 6.02 \times 10^{23}$ particles (the Avogadro constant).

Equation of State for an Ideal Gas

$$pV = nRT \qquad \text{or} \qquad pV = NkT$$

p = pressure (Pa)
V = volume (m^3)
n = number of moles (mol)
R = molar gas constant = $8.31\ \mathrm{J\,mol^{-1}K^{-1}}$
T = thermodynamic temperature (K)
N = number of molecules
k = Boltzmann constant = $1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$

The Boltzmann constant: $k = R/N_A$

Ideal Gas

An **ideal gas** is one that obeys $pV \propto T$ (where T is thermodynamic temperature) under all conditions. Real gases approximate ideal behaviour at low pressures and high temperatures.

Gas Laws — special cases of $pV = nRT$

- **Boyle's Law** (constant T): pV = constant, so $p_1V_1 = p_2V_2$
- **Charles' Law** (constant p): V/T = constant, so $V_1/T_1 = V_2/T_2$
- **Pressure Law** (constant V): p/T = constant, so $p_1/T_1 = p_2/T_2$

Common Mistake

Always use **thermodynamic temperature in kelvin** in gas law calculations — never degrees Celsius.

17 Kinetic Theory of Gases

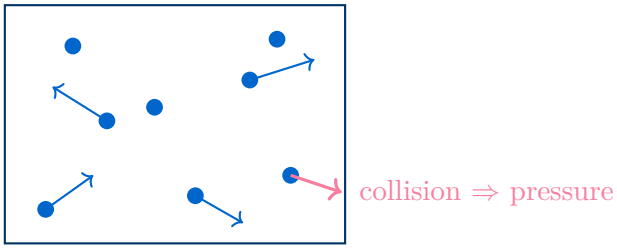
17.1 Assumptions of the Kinetic Model

Basic Assumptions of Kinetic Theory

1. The gas contains a **large number** of molecules moving in **random directions** with a range of speeds.
2. The molecules occupy **negligible volume** compared to the volume of the gas.
3. All collisions are **perfectly elastic** (no kinetic energy lost).
4. **Intermolecular forces are negligible** except during collisions.
5. The duration of collisions is **negligible** compared to the time between collisions.
6. Molecules obey **Newton's laws of motion**.

17.2 Pressure from Kinetic Theory

The pressure exerted by a gas arises from molecules colliding with the container walls. Each collision exerts a force; the average of many such collisions gives a steady pressure.



Random molecular motion causes pressure on walls

Kinetic Theory Equation

$$pV = \frac{1}{3}Nm\langle c^2\rangle$$

p	= pressure (Pa)
V	= volume (m^3)
N	= total number of molecules
m	= mass of one molecule (kg)
$\langle c^2\rangle$	= mean square speed ($m^2\,s^{-2}$)

The **root-mean-square speed**: $c_{\text{r.m.s.}} = \sqrt{\langle c^2\rangle}$

Note: $c_{\text{r.m.s.}}$ is not the same as the mean speed $\bar{c}$. Because we square first then average, faster molecules contribute more. In general $c_{\text{r.m.s.}} > \bar{c}$.

Derivation of $pV = \frac{1}{3}Nm\langle c^2\rangle$

Consider one molecule of mass m moving with speed c_x in the x-direction inside a box of side L.

Step 1 — change in momentum at one wall: The molecule hits the right wall and

bounces back elastically:

$$\Delta p = mc_x - (-mc_x) = 2mc_x$$

Step 2 — time between collisions with the same wall: The molecule must travel $2L$ before returning:

$$\Delta t = \frac{2L}{c_x}$$

Step 3 — force from one molecule:

$$F = \frac{\Delta p}{\Delta t} = \frac{2mc_x}{2L/c_x} = \frac{mc_x^2}{L}$$

Step 4 — pressure from N molecules: Summing over all N molecules and dividing by area L^2:

$$p = \frac{Nm\langle c_x^2\rangle}{L^3} = \frac{Nm\langle c_x^2\rangle}{V}$$

Step 5 — extend to 3 dimensions: By symmetry of random motion: $\langle c_x^2\rangle = \langle c_y^2\rangle = \langle c_z^2\rangle = \frac{1}{3}\langle c^2\rangle$

$$\boxed{pV = \tfrac{1}{3}Nm\langle c^2\rangle}$$

17.3 Molecular Kinetic Energy

Derivation of $\langle E_k\rangle = \frac{3}{2}kT$

Start with the two equations for an ideal gas:

$$pV = \tfrac{1}{3}Nm\langle c^2\rangle \qquad \text{and} \qquad pV = NkT$$

Equating the right-hand sides:

$$\tfrac{1}{3}Nm\langle c^2\rangle = NkT$$

Divide both sides by N and multiply by $\frac{3}{2}$:

$$\tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT$$

Since $\langle E_k\rangle = \frac{1}{2}m\langle c^2\rangle$:

$$\boxed{\langle E_k\rangle = \tfrac{3}{2}kT}$$

Average Translational Kinetic Energy

$$\langle E_k\rangle = \frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$$

The average kinetic energy of a molecule is **directly proportional to thermodynamic temperature**.

Key Deductions

- At absolute zero, molecules have minimum kinetic energy
- Molecules of different gases at the same temperature have the **same average KE**
- Heavier molecules move more **slowly** on average at the same temperature: $c_{\text{r.m.s.}} = \sqrt{3kT/m}$
- $c_{\text{r.m.s.}} \propto \sqrt{T}$ — doubling thermodynamic temperature increases r.m.s. speed by factor $\sqrt{2}$

18 Internal Energy

Internal Energy

The **internal energy** of a system is the sum of the **random kinetic energies** and **potential energies** of all the molecules in the system.

$$U = E_{k,\text{total}} + E_{p,\text{total}}$$

- Internal energy is a **function of state**.
- For an **ideal gas**: $E_p = 0$, so $U = E_{k,\text{total}}$ only.
- A rise in temperature $\Rightarrow$ increase in average KE $\Rightarrow$ increase in U.
- During a change of state: temperature (and KE) constant; E_p increases as bonds break; U increases.

19 The First Law of Thermodynamics

Work Done by/on a Gas

When a gas changes volume at **constant pressure**:

$$W = p\,\Delta V$$

W = work done **by** the gas (J)
p = pressure (Pa)
ΔV = change in volume (m^3)

First Law of Thermodynamics

$$\Delta U = q + W$$

ΔU = increase in internal energy of the system (J)
q = energy supplied **to** the system by heating (J)
W = work done **on** the system (J)

Sign Convention

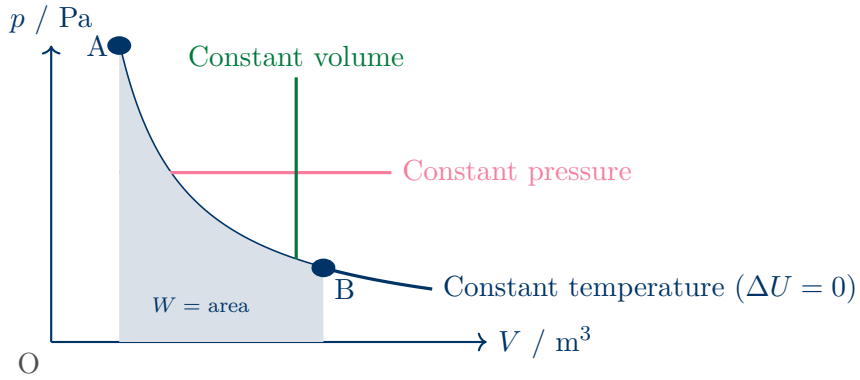
CIE uses $\Delta U = q + W$ where W is work done **on** the system. Some textbooks use $\Delta U = q - W$ where W is work done **by** the system. Always check which convention is being used.

Special Cases of the First Law

Process	Condition	Result
Constant temperature	$\Delta T = 0 \Rightarrow \Delta U = 0$	$q = -W$
No heat transfer	$q = 0$	$\Delta U = W$
Constant volume	$\Delta V = 0 \Rightarrow W = 0$	$\Delta U = q$
Constant pressure	p constant	$\Delta U = q - p\Delta V$

19.1 p–V Diagrams

The **area under a p–V curve** equals the work done by the gas during that process.



20 Formula Summary

Formula	Quantity	Units
$T/\mathrm{K} = \theta/°\mathrm{C} + 273.15$	Temperature conversion	K
$Q = mc\Delta T$	Sensible heat	J
$Q = mL$	Latent heat	J
$pV = nRT$	Ideal gas (moles)	Pa, m^3, K
$pV = NkT$	Ideal gas (molecules)	Pa, m^3, K
$k = R/N_A$	Boltzmann constant	$\mathrm{J\,K^{-1}}$
$pV = \frac{1}{3}Nm\langle c^2\rangle$	Kinetic theory	Pa
$\langle E_k\rangle = \frac{3}{2}kT$	Mean molecular KE	J
$W = p\Delta V$	Work done by gas	J
$\Delta U = q + W$	First law	J

Constants: $R = 8.31\ \mathrm{J\,mol^{-1}K^{-1}}$ $k = 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$ $N_A = 6.02 \times 10^{23}\ \mathrm{mol^{-1}}$

21 Worked Examples

Example 1 — Specific Heat Capacity

Question: A 2.0 kg block of aluminium ($c = 900\ \mathrm{J\,kg^{-1}K^{-1}}$) is heated from 20 °C to 80 °C. Calculate the energy supplied.

Solution

$$Q = mc\Delta T = 2.0 \times 900 \times (80 - 20) = \mathbf{1.08 \times 10^5}\ \mathrm{J}$$

Example 2 — Ideal Gas Law

Question: A fixed mass of ideal gas has pressure 1.2×10^5 Pa and volume $3.0 \times 10^{-3}\ \mathrm{m}^3$ at 27 °C. It is heated at constant pressure until its volume doubles. Find the final temperature.

Solution

$T_1 = 27 + 273.15 = 300.15$ K. At constant pressure, $V \propto T$:

$$T_2 = T_1 \times \frac{V_2}{V_1} = 300.15 \times 2 = \mathbf{600}\ \mathrm{K}\ (\mathbf{327}\ °\mathrm{C})$$

Example 3 — r.m.s. Speed

Question: Calculate the r.m.s. speed of nitrogen molecules ($M = 0.028\ \mathrm{kg\,mol^{-1}}$) at 300 K.

Solution

Mass of one molecule: $m = 0.028/6.02 \times 10^{23} = 4.65 \times 10^{-26}$ kg

$$c_{\text{r.m.s.}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times 300}{4.65 \times 10^{-26}}} = \sqrt{2.67 \times 10^5} = \mathbf{517}\ \mathrm{m\,s^{-1}}$$

Example 4 — First Law

Question: A gas is compressed. 650 J of work is done on the gas and 200 J of heat flows out. Find the change in internal energy.

Solution

$W = +650$ J (work done *on* gas), $q = -200$ J (heat leaves)

$$\Delta U = q + W = -200 + 650 = +\mathbf{450}\ \mathrm{J}$$

Internal energy **increases** by 450 J.

22 Practice Exam Questions

Section A — Short Answer Questions

Q1. Define *specific latent heat of vaporisation* and explain why it is greater than the specific latent heat of fusion for the same substance.

[3 marks]

Q2. State the basic assumptions of the kinetic theory of gases.

[4 marks]

Q3. Explain what is meant by *internal energy* and state how it differs for an ideal gas compared to a real gas.

[3 marks]

Q4. A gas at pressure 2.0×10^5 Pa and temperature 300 K occupies a volume of 0.40 m^3. Calculate the number of molecules present.

[2 marks]

Q5. Show that the mean kinetic energy of a gas molecule is $\frac{3}{2}kT$ by combining the kinetic theory equation with the ideal gas equation.

[3 marks]

Section B — Longer Structured Questions

Q6. A sample of water of mass 0.50 kg is heated from 20 °C to 100 °C and then completely vaporised.
($c_{\text{water}} = 4200\ \text{J kg}^{-1}\text{K}^{-1}$, $L_v = 2.26 \times 10^6\ \text{J kg}^{-1}$)

(a) Calculate the energy needed to heat the water from 20 °C to 100 °C.

[2 marks]

(b) Calculate the energy needed to vaporise the water at 100 °C.

[2 marks]

(c) Explain in terms of molecular behaviour why energy is needed to vaporise water even though the temperature does not change.

[2 marks]

Q7. A fixed mass of ideal gas undergoes the cycle A → B → C → A on a p–V diagram where A → B is constant temperature expansion, B → C is constant volume pressure decrease, and C → A is constant pressure compression.

(a) State the change in internal energy during A → B and justify your answer.
[2 marks]

(b) Using the first law, find q for A → B given that the work done by the gas is 440 J.
[2 marks]

(c) Describe the energy changes during the constant volume process B → C.
[2 marks]

23 Mark Scheme and Answers

Q1. Specific latent heat of vaporisation is the energy per unit mass required to change a substance from liquid to gas at constant temperature [1]; greater than L_f because molecules must be completely separated, requiring work against intermolecular forces over a much greater distance than in melting [2].

Q2. Large number of molecules moving randomly in all directions with a range of speeds [1]; occupying negligible volume compared to container [1]; collisions perfectly elastic [1]; intermolecular forces negligible except during collisions; collisions of negligible duration; obey Newton's laws [1].

Q3. Internal energy is the sum of random kinetic and potential energies of all molecules [1]; for an ideal gas, $E_p = 0$ so internal energy is kinetic energy only [1]; for a real gas, molecules have potential energy due to intermolecular forces [1].

Q4. $N = pV/kT = (2.0 \times 10^5 \times 0.40)/(1.38 \times 10^{-23} \times 300) = \mathbf{1.93 \times 10^{25}}$ molecules [2].

Q5. $pV = \frac{1}{3}Nm\langle c^2\rangle$ [1]; equate with $pV = NkT$: $\frac{1}{3}m\langle c^2\rangle = kT$ [1]; so $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$ [1].

Q6(a). $Q = 0.50 \times 4200 \times 80 = \mathbf{1.68 \times 10^5}$ J [2].

Q6(b). $Q = 0.50 \times 2.26 \times 10^6 = \mathbf{1.13 \times 10^6}$ J [2].

Q6(c). KE of molecules unchanged (temperature constant) [1]; energy breaks intermolecular bonds as molecules separate completely, increasing potential energy [1].

Q7(a). $\Delta U = 0$ [1]; temperature is constant and for an ideal gas internal energy depends only on temperature [1].

Q7(b). $\Delta U = 0$, work done on gas $= -440$ J; $q = +440$ J — heat flows into gas [2].

Q7(c). No work done ($\Delta V = 0$) [1]; temperature and KE decrease; heat flows out; internal energy decreases [1].

24 Revision Checklist

	Learning Objective	Confidence (1–3)
☐	Explain thermal equilibrium and direction of heat flow	
☐	Convert between Celsius and Kelvin	
☐	State examples of thermometric properties	
☐	Define and use specific heat capacity: $Q = mc\Delta T$	
☐	Define and use specific latent heat: $Q = mL$	
☐	State the assumptions of kinetic theory	
☐	Use $pV = nRT$ and $pV = NkT$	
☐	Derive and use $pV = \frac{1}{3}Nm\langle c^2\rangle$	
☐	Show that mean KE of a molecule is $\frac{3}{2}kT$	
☐	Define internal energy and relate to temperature	
☐	Apply the first law $\Delta U = q + W$	
☐	Interpret and use p–V diagrams	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

Remember: understanding *why* formulas work is more powerful than memorising them. Practice drawing p–V diagrams and deriving key results from first principles.

CIE International A Level Physics

Topic 17

Oscillations

Revision Booklet

This booklet covers:

- Simple Harmonic Motion
- Displacement, Velocity & Acceleration
- Energy in SHM
- Damped Oscillations
- Forced Oscillations & Resonance

25	Core Concepts and Definitions	35
26	Equations of Simple Harmonic Motion	35
27	Energy in Simple Harmonic Motion	36
28	Damped and Forced Oscillations	38
29	Formula Summary Sheet	40
30	Worked Examples	40
31	Practice Exam Questions	41
32	Mark Scheme and Answers	44
33	Revision Checklist	46

25 Core Concepts and Definitions

Oscillatory Motion — Key Terms

- **Displacement** x: the distance from the equilibrium position, with direction (m).
- **Amplitude** x_0: the maximum displacement from equilibrium (m).
- **Period** T: the time for one complete oscillation (s).
- **Frequency** f: the number of complete oscillations per second (Hz); $f = 1/T$.
- **Angular frequency** ω: $\omega = 2\pi f = 2\pi/T$ (rad s^{-1}).
- **Phase difference** ϕ: the fraction of a cycle by which one oscillation leads or lags another, expressed in radians.

Simple Harmonic Motion (SHM)

An oscillation is **simple harmonic** if the acceleration of the object is:

- **proportional** to its displacement from a fixed equilibrium point, and
- always directed **towards** that equilibrium point (opposite to displacement).

$$a \propto -x$$

26 Equations of Simple Harmonic Motion

Defining Equation of SHM

$$a = -\omega^2 x$$

a = acceleration (m s^{-2})
ω = angular frequency (rad s^{-1})
x = displacement from equilibrium (m)

The negative sign confirms acceleration is always directed **opposite** to displacement.

Displacement; Velocity and Acceleration

$$x = x_0 \sin \omega t \qquad \text{(starting from equilibrium)}$$

$$v = v_0 \cos \omega t \qquad \text{where } v_0 = \omega x_0$$

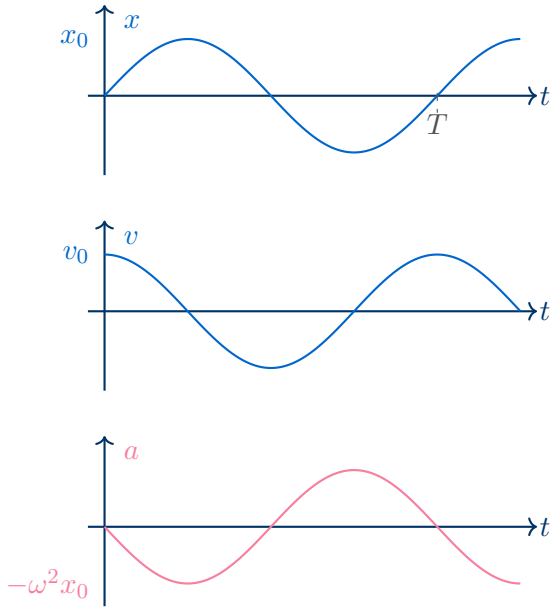
$$v = \pm\, \omega \sqrt{x_0^2 - x^2}$$

x_0 = amplitude (m)
v_0 = maximum speed = ωx_0 (m s^{-1})
t = time (s)

Maxima and Minima

- Acceleration is **maximum** at maximum displacement ($x = \pm x_0$): $|a_{\text{max}}| = \omega^2 x_0$
- Speed is **maximum** at the equilibrium position ($x = 0$): $v_{\text{max}} = \omega x_0$
- Speed is **zero** at the turning points ($x = \pm x_0$).

Graphs of x, v and a against time



Phase Relationships

- v leads x by $\frac{\pi}{2}$ rad (quarter period ahead).
- a leads v by $\frac{\pi}{2}$ rad, so a is **antiphase** (π rad ahead) with x.

Common Mistake

The equation $x = x_0 \sin \omega t$ assumes the object starts at the **equilibrium position** at $t = 0$. If it starts at maximum displacement, use $x = x_0 \cos \omega t$ instead. Always check the initial conditions.

27 Energy in Simple Harmonic Motion

Total Energy of a SHM System

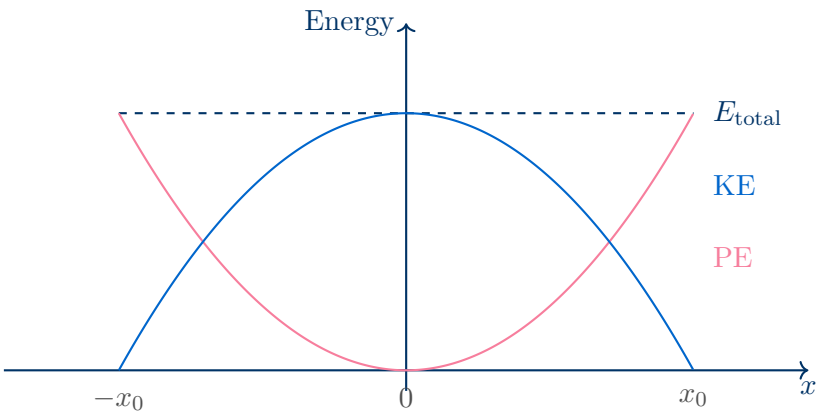
$$E = \tfrac{1}{2} m \omega^2 x_0^2$$

E = total mechanical energy (J)
m = mass of the oscillating object (kg)
ω = angular frequency (rad s^{-1})
x_0 = amplitude (m)

Energy Interchange During SHM

- At the **equilibrium position** ($x = 0$): KE is maximum, PE is zero.
- At the **turning points** ($x = \pm x_0$): KE is zero, PE is maximum.
- The **total energy remains constant** throughout (assuming no damping).
- $\text{KE} = \frac{1}{2}m\omega^2(x_0^2 - x^2)$ $\qquad$ $\text{PE} = \frac{1}{2}m\omega^2 x^2$

Energy–displacement graph



28 Damped and Forced Oscillations

28.1 Damping

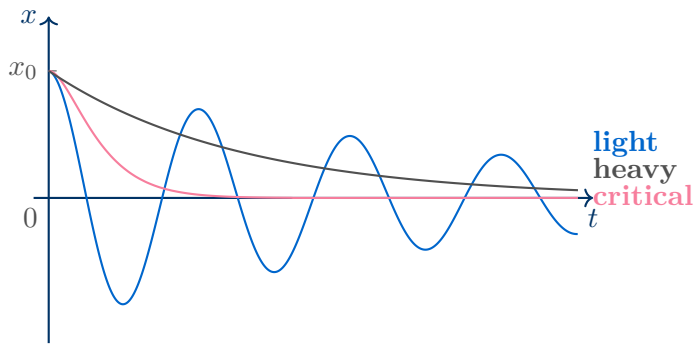
Damping

Damping occurs when a resistive force (e.g., air resistance, viscosity) acts on an oscillating system. Energy is removed from the system, causing the amplitude to **decrease over time**. The frequency is approximately unchanged for light damping.

Types of Damping

- **Light damping**: amplitude decreases gradually over many oscillations; the system oscillates for a long time before coming to rest.
- **Critical damping**: the system returns to equilibrium in the **shortest possible time** without oscillating. Used in car suspension and door closers.
- **Heavy (overdamping)**: the system returns to equilibrium **slowly** without oscillating; slower return than critical damping.

Displacement–time graphs for the three types of damping



28.2 Forced Oscillations and Resonance

Forced Oscillations and Natural Frequency

- Every oscillating system has a **natural frequency** f_0 at which it oscillates freely when displaced and released.
- A **forced oscillation** occurs when a periodic **driving force** is applied to the system at a **driving frequency** f.
- The system oscillates at the **driving frequency**, not necessarily at its natural frequency.

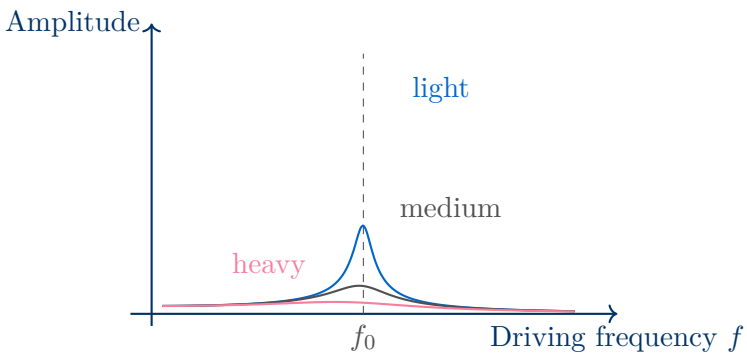
Resonance

Resonance occurs when the driving frequency equals the natural frequency of the system ($f = f_0$). At resonance:

- The amplitude of oscillation is a **maximum**.
- Energy transfer from the driver to the system is most efficient.

- The degree of damping determines how sharp the resonance peak is and the maximum amplitude reached.

Amplitude–frequency graph (resonance curves)



Effect of Damping on Resonance

- More damping ⇒ lower and broader resonance peak.
- More damping ⇒ peak shifts **below** f_0 (towards lower frequencies).
- Less damping ⇒ sharper, taller peak with maximum amplitude closer to f_0.
- With no damping the peak would be infinite at exactly f_0.

Resonance in Real Life

Resonance can be **useful** (e.g. MRI scanners, musical instruments, microwave ovens) or **destructive** (e.g. bridges vibrating in wind, buildings in earthquakes). Engineers use damping to control unwanted resonance.

29 Formula Summary Sheet

Formula	Quantity	Units
$a = -\omega^2 x$	SHM defining equation	m s^{-2}
$\omega = 2\pi f = \dfrac{2\pi}{T}$	Angular frequency	rad s^{-1}
$x = x_0 \sin \omega t$	Displacement (from equilibrium at $t = 0$)	m
$v = v_0 \cos \omega t$	Velocity	m s^{-1}
$v = \pm\omega\sqrt{x_0^2 - x^2}$	Speed at displacement x	m s^{-1}
$v_0 = \omega x_0$	Maximum speed	m s^{-1}
$a_{\text{max}} = \omega^2 x_0$	Maximum acceleration	m s^{-2}
$E = \frac{1}{2}m\omega^2 x_0^2$	Total energy of SHM system	J
$\text{KE} = \frac{1}{2}m\omega^2(x_0^2 - x^2)$	Kinetic energy at displacement x	J
$\text{PE} = \frac{1}{2}m\omega^2 x^2$	Potential energy at displacement x	J

Key relationships: $T = 1/f$, $\omega = 2\pi f$, $v_{\text{max}} = \omega x_0$, $a_{\text{max}} = \omega^2 x_0$
Phase: v leads x by $\pi/2$; a is antiphase with x (leads by π)

30 Worked Examples

Example 1 — Finding Angular Frequency and Max Speed

Question: A mass oscillates with SHM, amplitude 4.0 cm and period 0.80 s. Calculate (a) the angular frequency and (b) the maximum speed.

Solution

(a) $\omega = \dfrac{2\pi}{T} = \dfrac{2\pi}{0.80} = \mathbf{7.85}$ rad s^{-1}

(b) $v_{\text{max}} = \omega x_0 = 7.85 \times 0.040 = \mathbf{0.31}$ m s^{-1}

Example 2 — Speed at a Given Displacement

Question: The same mass ($\omega = 7.85$ rad s^{-1}, $x_0 = 4.0$ cm) has displacement $x = 2.5$ cm. Find its speed.

Solution

$$v = \omega\sqrt{x_0^2 - x^2} = 7.85 \times \sqrt{(0.040)^2 - (0.025)^2}$$

$$v = 7.85 \times \sqrt{1.60 \times 10^{-3} - 6.25 \times 10^{-4}} = 7.85 \times \sqrt{9.75 \times 10^{-4}}$$

$$v = 7.85 \times 0.0312 = \mathbf{0.245}\ \text{m s}^{-\mathbf{1}}$$

Example 3 — Total Energy of SHM System

Question: A 0.20 kg mass oscillates with $\omega = 7.85$ rad s^{-1} and amplitude 4.0 cm. Calculate the total energy.

Solution

$$E = \tfrac{1}{2}m\omega^2 x_0^2 = \tfrac{1}{2} \times 0.20 \times (7.85)^2 \times (0.040)^2$$

$$E = 0.10 \times 61.6 \times 1.60 \times 10^{-3} = \mathbf{9.9 \times 10^{-3}}\ \text{J}$$

31 Practice Exam Questions

Section A — Short Answer Questions

Q1. State the two conditions that define simple harmonic motion.

[2 marks]

Q2. Define the terms *amplitude*, *period* and *angular frequency* for an oscillating system.

[3 marks]

Q3. An object undergoes SHM with frequency 2.5 Hz and amplitude 6.0 cm.

(a) Calculate the angular frequency. *[1 mark]*

(b) Calculate the maximum acceleration. *[2 marks]*

(c) Calculate the speed when $x = 4.0$ cm. *[2 marks]*

Q4. Sketch displacement–time graphs for *light*, *critical* and *heavy* damping on the same axes, starting from the same initial displacement. Label each curve.

[3 marks]

Q5. Explain what is meant by *resonance* and state the condition under which it occurs.

[2 marks]

Section B — Longer Structured Questions

Q6. A 0.15 kg mass is attached to a spring and oscillates vertically with SHM. Its displacement is given by $x = 0.050\sin(12t)$, where x is in metres and t in seconds.

(a) Write down the amplitude and angular frequency of the motion.

[2 marks]

(b) Show that the maximum speed is 0.60 m s^{-1}.

[2 marks]

(c) Calculate the total energy of the oscillation.

[2 marks]

(d) Sketch graphs on the same axes showing how the kinetic energy and potential energy vary with displacement x. Label your axes clearly.

[3 marks]

Q7. A child on a swing is pushed periodically by a parent.

(a) Explain why the amplitude of the swing increases when the pushing frequency equals the natural frequency of the swing.

[2 marks]

(b) The parent now pushes at a frequency higher than the natural frequency. Describe and explain what happens to the amplitude.

[2 marks]

(c) Air resistance acts on the swing. Describe how this affects the resonance curve compared to an undamped system.

[2 marks]

32 Mark Scheme and Answers

Q1. Acceleration is proportional to displacement from a fixed (equilibrium) point [1]; acceleration is always directed towards that point / opposite to displacement [1].

Q2. Amplitude: maximum displacement from equilibrium [1]. Period: time for one complete oscillation [1]. Angular frequency: $\omega = 2\pi f = 2\pi/T$ (rad s^{-1}) [1].

Q3(a). $\omega = 2\pi f = 2\pi \times 2.5 = \mathbf{15.7}$ rad s^{-1} [1].

Q3(b). $a_{\max} = \omega^2 x_0 = (15.7)^2 \times 0.060$ [1] $= \mathbf{14.8}$ m s^{-2} [1].

Q3(c). $v = \omega\sqrt{x_0^2 - x^2} = 15.7 \times \sqrt{(0.060)^2 - (0.040)^2}$ [1] $= 15.7 \times 0.0447 = \mathbf{0.70}$ m s^{-1} [1].

Q4. All three start from same x_0 [1]; light damping: decaying sinusoid, multiple oscillations; critical: returns to zero smoothly in shortest time without oscillating; heavy: slower return to zero, no oscillation — all three correctly labelled [2].

Q5. Resonance: when the driving frequency equals the natural frequency of the system [1]; the amplitude of oscillation reaches a maximum [1].

Q6(a). Amplitude $x_0 = 0.050$ m [1]; angular frequency $\omega = 12$ rad s^{-1} [1].

Q6(b). $v_{\text{max}} = \omega x_0 = 12 \times 0.050 = 0.60$ m s^{-1} [2] (must show substitution).

Q6(c). $E = \frac{1}{2}m\omega^2 x_0^2 = \frac{1}{2} \times 0.15 \times 144 \times 2.5 \times 10^{-3}$ [1] $= \mathbf{2.7 \times 10^{-2}}$ J [1].

Q6(d). PE: upward parabola, zero at $x = 0$, maximum E at $x = \pm x_0$ [1]; KE: inverted parabola, maximum E at $x = 0$, zero at $x = \pm x_0$ [1]; axes correctly labelled with E, $-x_0$, 0, x_0 [1].

Q7(a). Energy is transferred most efficiently from driver to system when $f_{\text{drive}} = f_0$ [1]; each push is in phase with the motion so energy is added each cycle, increasing amplitude [1].

Q7(b). Amplitude decreases [1]; driving frequency is not matched to natural frequency so energy transfer is less efficient / pushes are out of phase with motion [1].

Q7(c). The resonance peak is lower (smaller maximum amplitude) [1]; the peak is broader and shifts slightly to a frequency below f_0 [1].

33 Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

	Learning Objective	Confidence (1–3)
☐	Define displacement, amplitude, period, frequency and angular frequency	
☐	State and apply the two conditions for SHM	
☐	Use $a = -\omega^2 x$ to identify and analyse SHM	
☐	Use $x = x_0 \sin \omega t$ and $v = v_0 \cos \omega t$	
☐	Use $v = \pm\omega\sqrt{x_0^2 - x^2}$ to find speed at any displacement	
☐	Sketch and interpret graphs of x, v and a against t	
☐	Describe the phase relationships between x, v and a	
☐	Describe the interchange between KE and PE during SHM	
☐	Use $E = \frac{1}{2}m\omega^2 x_0^2$ for total energy	
☐	Sketch and interpret KE and PE against displacement graphs	
☐	Explain and distinguish light, critical and heavy damping	
☐	Sketch displacement–time graphs for the three types of damping	
☐	Explain resonance and the condition for it to occur	
☐	Describe the effect of damping on the resonance curve	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

Remember: in SHM, acceleration and displacement are always linked by $a = -\omega^2 x$. If you can start from this equation and derive everything else, you truly understand the topic.

CIE International A Level Physics

Topic 18

Electric Fields

Revision Booklet

This booklet covers:

- Electric Fields and Field Lines
- Uniform Electric Fields
- Coulomb's Law
- Electric Field of a Point Charge
- Electric Potential

34	Core Concepts and Definitions	49
35	Uniform Electric Fields	49
36	Coulomb's Law	50
37	Electric Field of a Point Charge	51
38	Electric Potential	51
39	Equipotential Surfaces	53
40	Formula Summary Sheet	54
41	Worked Examples	54
42	Practice Exam Questions	55
43	Mark Scheme and Answers	59
44	Revision Checklist	60

34 Core Concepts and Definitions

Electric Field

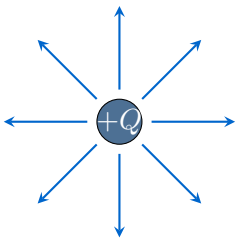
An **electric field** is a region of space in which a charged object experiences a force.

- Electric field is defined as the **force per unit positive charge** acting on a small stationary test charge placed at that point.
- It is a **vector** quantity, directed along the force on a positive test charge.
- Units: N $C^{-1} \equiv$ V m^{-1}.

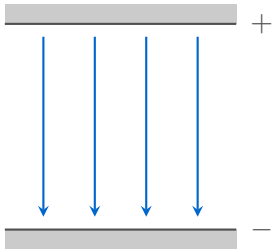
$$E = \frac{F}{q}$$

Radial vs Uniform Fields

- **Radial field** (around a point charge or sphere): field lines point radially inward (negative charge) or outward (positive charge); E decreases with distance.
- **Uniform field** (between parallel plates): field lines are parallel and equally spaced; E is constant throughout.



Radial Field (+)



Uniform Field

35 Uniform Electric Fields

Field Strength Between Parallel Plates

$$E = \frac{\Delta V}{\Delta d}$$

E = electric field strength (V m^{-1})
ΔV = potential difference between the plates (V)
Δd = separation of the plates (m)

Motion of Charged Particles in a Uniform Field

- A charge q in a uniform field E experiences a constant force $F = qE$.
- The motion is analogous to projectile motion in a gravitational field:
  - Along the field: **uniform acceleration** $a = qE/m$.

  - Perpendicular to the field: **constant velocity** (if no other forces).
- The path of the charge is **parabolic**.

Common Mistake

$E = \Delta V/\Delta d$ applies **only** to uniform fields (parallel plates). Do not apply it to the field around a point charge — use $E = Q/(4\pi\varepsilon_0 r^2)$ instead.

36 Coulomb's Law

Coulomb's Law

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$$

F	= electrostatic force between the charges (N)
Q_1, Q_2	= the two point charges (C)
r	= separation between the charges (m)
ε_0	= permittivity of free space = 8.85×10^{-12} F m^{-1}

Key Points

- Like charges **repel**; unlike charges **attract**.
- The force obeys an **inverse-square law**: double the distance, quarter the force.
- Applies strictly to **point charges**, and to uniform spheres (treat as point charge at centre).
- Compare with gravity: same inverse-square form but gravity is **always attractive**.

Comparison: Coulomb's Law vs Newton's Law of Gravitation

Coulomb's Law	**Newton's Law**
$F = \dfrac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$	$F = \dfrac{G m_1 m_2}{r^2}$
Can be attractive or repulsive	Always attractive
Acts between charges	Acts between masses
Much stronger force	Much weaker force

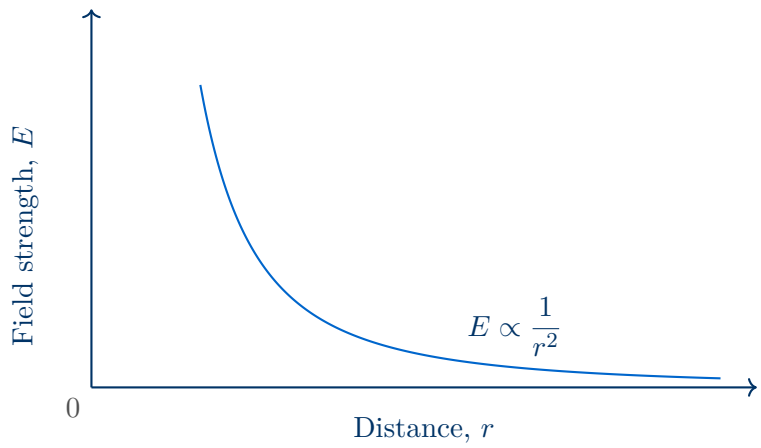
37 Electric Field of a Point Charge

Electric Field Strength due to a Point Charge

$$E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

E = electric field strength at distance r (N C^{-1})
Q = point charge creating the field (C)
r = distance from the centre of Q (m)
ε_0 = 8.85×10^{-12} F m^{-1}

Variation of E with distance r



Comparing E and g fields

- Both follow an inverse-square law with distance.
- $E = Q/(4\pi\varepsilon_0 r^2)$ parallels $g = GM/r^2$.
- $F = qE$ parallels $F = mg$.
- Unlike gravitational fields, electric fields can point inward or outward depending on the sign of Q.

38 Electric Potential

Definition of Electric Potential V

The **electric potential** at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point.

$$V = \frac{W}{q} \qquad \text{units: J C}^{-1} \equiv \text{V}$$

- $V = 0$ at infinity (by convention).
- Around a positive charge: $V > 0$ (work must be done *against* repulsion).

- Around a negative charge: $V < 0$ (work is done *by* the field as the test charge moves in).

Electric Potential due to a Point Charge

$$V = \frac{Q}{4\pi\varepsilon_0 r}$$

V = electric potential at distance r (V)
Q = point charge (C)
r = distance from the charge (m)

38.1 Relationship Between E and V

Field Strength from Potential Gradient

$$E = -\frac{\Delta V}{\Delta r}$$

The electric field strength equals the **negative gradient** of the potential–distance graph. The area under an E–r graph gives the change in potential ΔV.

38.2 Electric Potential Energy

Potential Energy of Two Point Charges

$$E_P = \frac{Qq}{4\pi\varepsilon_0 r}$$

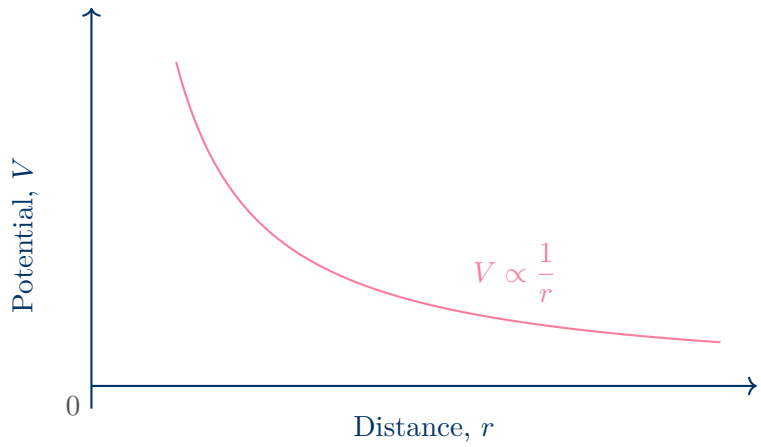
E_P = electric potential energy (J)
Q, q = the two point charges (C)
r = separation (m)

Note: $E_P = qV$, where V is the potential due to charge Q at the location of q.

Signs of Potential Energy

- Like charges ($Qq > 0$): $E_P > 0$ — energy must be supplied to bring them together.
- Unlike charges ($Qq < 0$): $E_P < 0$ — energy is released as they come together.
- $E_P = 0$ at infinite separation.

Graphs of V and E against r for a positive point charge



Common Mistake

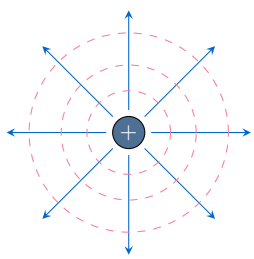
Note the difference in distance dependence: $E \propto 1/r^2$ but $V \propto 1/r$. Students often mix these up. Remember: potential falls off more *slowly* than field strength with distance.

39 Equipotential Surfaces

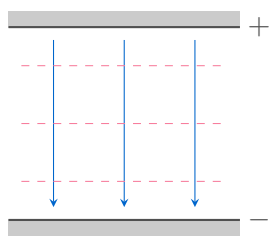
Equipotentials

An **equipotential surface** is a surface on which the electric potential is the same at every point.

- No work is done moving a charge *along* an equipotential.
- Equipotentials are always **perpendicular** to field lines.
- Around a point charge: equipotentials are concentric spheres.
- Between parallel plates: equipotentials are parallel planes equally spaced (for uniform field).



Point charge



Parallel plates

40 Formula Summary Sheet

Formula	Quantity	Units
$E = \dfrac{F}{q}$	Electric field strength (definition)	N C^{-1}
$F = qE$	Force on a charge in a field	N
$E = \dfrac{\Delta V}{\Delta d}$	Field between parallel plates (uniform)	V m^{-1}
$F = \dfrac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$	Coulomb's Law	N
$E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$	Field due to a point charge	N C^{-1}
$V = \dfrac{Q}{4\pi\varepsilon_0 r}$	Potential due to a point charge	V
$E = -\dfrac{\Delta V}{\Delta r}$	Field from potential gradient	V m^{-1}
$E_P = \dfrac{Qq}{4\pi\varepsilon_0 r}$	Electric potential energy	J
$E_P = qV$	Potential energy of charge in field	J

Constants: $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}, $\dfrac{1}{4\pi\varepsilon_0} = 8.99\times10^{9}$ N m^{2}C^{-2}, $e = 1.60\times10^{-19}$ C

Note: $E \propto 1/r^2$ (inverse-square) but $V \propto 1/r$ (inverse).

41 Worked Examples

Example 1 — Force Between Two Charges

Question: Calculate the electrostatic force between two charges of $+3.0\ \mu$C and $-5.0\ \mu$C separated by 0.12 m.

Solution

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2} = \frac{3.0\times10^{-6}\times5.0\times10^{-6}}{4\pi\times8.85\times10^{-12}\times(0.12)^2}$$

$$F = \frac{1.5\times10^{-11}}{1.61\times10^{-12}} = \mathbf{9.3}\ \text{N}$$

The force is **attractive** (unlike charges).

Example 2 — Field Strength and Potential at a Distance

Question: A point charge $Q = +4.0\ \mu$C. Calculate (a) the electric field strength and (b) the electric potential at a distance of 0.30 m.

Solution

(a) $$E = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{4.0 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times (0.30)^2}$$

$$E = \frac{4.0 \times 10^{-6}}{1.005 \times 10^{-10}} \times (0.09)^{-1}$$

$$E = \frac{4.0 \times 10^{-6}}{1.005 \times 10^{-10}} = \mathbf{4.0 \times 10^5}\ \text{N C}^{-1}$$

(b) $$V = \frac{Q}{4\pi\varepsilon_0 r} = \frac{4.0 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times 0.30} = \mathbf{1.2 \times 10^5}\ \text{V}$$

Example 3 — Uniform Field Between Plates

Question: Two parallel plates are separated by 4.0 mm with a potential difference of 240 V. Calculate (a) the field strength and (b) the force on an electron between the plates.

Solution

(a) $$E = \frac{\Delta V}{\Delta d} = \frac{240}{4.0 \times 10^{-3}} = \mathbf{6.0 \times 10^4}\ \text{V m}^{-1}$$

(b) $$F = qE = 1.6 \times 10^{-19} \times 6.0 \times 10^4 = \mathbf{9.6 \times 10^{-15}}\ \text{N}$$

42 Practice Exam Questions

Section A — Short Answer Questions

Q1. Define electric field strength and state its SI units.

[2 marks]

Q2. State Coulomb's Law and identify one similarity and one difference compared with Newton's Law of Gravitation.

[3 marks]

Q3. Define electric potential at a point. Explain why the potential around an isolated positive charge is positive, but around an isolated negative charge is negative.

[3 marks]

Q4. Two parallel plates are 6.0 mm apart. The electric field strength between them is 5.0×10^4 V m^{-1}. Calculate the potential difference between the plates.

[2 marks]

Q5. A point charge produces an electric field of 3.6×10^5 N C^{-1} at a distance of 0.10 m. Calculate the magnitude of the charge.

[3 marks]

Section B — Longer Structured Questions

Q6. Two point charges $Q_1 = +6.0\ \mu$C and $Q_2 = +6.0\ \mu$C are placed 0.20 m apart.

(a) Calculate the force between the two charges.

[2 marks]

(b) Calculate the electric field strength at the midpoint between the two charges. Explain your reasoning.

[3 marks]

(c) Calculate the electric potential at the midpoint between the two charges.

[3 marks]

Q7. An electron (mass 9.11×10^{-31} kg, charge -1.6×10^{-19} C) enters horizontally between two parallel plates. The plates are 30 mm apart and have a potential difference of 150 V. The electron enters midway between the plates with horizontal speed 4.0×10^{7} m s^{-1}.

(a) Calculate the electric field strength between the plates.

[1 mark]

(b) Calculate the vertical acceleration of the electron.

[2 marks]

(c) The plates are 60 mm long. Calculate the vertical deflection of the electron as it exits the plates.

[3 marks]

(d) State and explain whether the electron hits one of the plates before exiting.

[2 marks]

43 Mark Scheme and Answers

Q1. Electric field strength is the force per unit positive charge acting on a small stationary test charge placed at that point [1]; units: N C^{-1} or V m^{-1} [1].

Q2. The force between two point charges is proportional to the product of the charges and inversely proportional to the square of their separation [1]. Similarity: both follow an inverse-square law [1]. Difference: gravity is always attractive; electrostatic force can be attractive or repulsive [1].

Q3. Electric potential at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point [1]. Around a positive charge: the test charge is repelled, so work must be done *on* it to bring it in from infinity — the potential is positive [1]. Around a negative charge: the test charge is attracted, so work is done *by* the field — the potential is negative [1].

Q4. $\Delta V = E \times \Delta d = 5.0 \times 10^4 \times 6.0 \times 10^{-3} = \mathbf{300}$ V [2].

Q5. $E = Q/(4\pi\varepsilon_0 r^2)$ [1]; $Q = E \times 4\pi\varepsilon_0 r^2 = 3.6 \times 10^5 \times 4\pi \times 8.85 \times 10^{-12} \times (0.10)^2$ [1] $= \mathbf{4.0 \times 10^{-7}}$ C [1].

Q6(a). $F = \dfrac{Q_1 Q_2}{4\pi\varepsilon_0 r^2} = \dfrac{(6.0 \times 10^{-6})^2}{4\pi \times 8.85 \times 10^{-12} \times (0.20)^2}$ [1] $= \mathbf{8.1}$ N [1].

Q6(b). By symmetry, the two equal charges produce equal and opposite field contributions at the midpoint [1]; the fields cancel, so $E = \mathbf{0}$ at the midpoint [2].

Q6(c). $V_1 = V_2 = \dfrac{Q}{4\pi\varepsilon_0 r} = \dfrac{6.0 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times 0.10}$ [1] $= 5.4 \times 10^5$ V; total $V = V_1 + V_2 = \mathbf{1.08 \times 10^6}$ V [2].

Q7(a). $E = \Delta V/\Delta d = 150/(30 \times 10^{-3}) = \mathbf{5.0 \times 10^3}$ V m^{-1} [1].

Q7(b). $F = qE = 1.6 \times 10^{-19} \times 5.0 \times 10^3 = 8.0 \times 10^{-16}$ N [1]; $a = F/m = 8.0 \times 10^{-16}/(9.11 \times 10^{-31}) = \mathbf{8.8 \times 10^{14}}$ m s^{-2} [1].

Q7(c). Time in field: $t = L/v = 0.060/(4.0 \times 10^7) = 1.5 \times 10^{-9}$ s [1]; vertical deflection: $y = \frac{1}{2}at^2 = \frac{1}{2} \times 8.8 \times 10^{14} \times (1.5 \times 10^{-9})^2$ [1] $= \mathbf{9.9 \times 10^{-4}}$ m $\approx \mathbf{1.0}$ mm [1].

Q7(d). The electron enters midway so has 15 mm to the nearest plate [1]; deflection of ≈ 1 mm $\ll 15$ mm, so the electron **does not** hit a plate [1].

44 Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

	Learning Objective	Confidence (1–3)
☐	Define electric field strength as force per unit positive charge; use $E = F/q$	
☐	Represent electric fields using field lines for point charges and parallel plates	
☐	Use $E = \Delta V/\Delta d$ for a uniform field between parallel plates	
☐	Describe the parabolic motion of a charge in a uniform electric field	
☐	State and apply Coulomb's Law $F = Q_1Q_2/(4\pi\varepsilon_0 r^2)$	
☐	Use $E = Q/(4\pi\varepsilon_0 r^2)$ for the field due to a point charge	
☐	Define electric potential; explain why sign depends on sign of source charge	
☐	Use $V = Q/(4\pi\varepsilon_0 r)$ for potential due to a point charge	
☐	Apply $E = -\Delta V/\Delta r$ to relate field strength and potential gradient	
☐	Use $E_P = Qq/(4\pi\varepsilon_0 r)$ for potential energy of two charges	
☐	Sketch and interpret equipotential diagrams for point charges and parallel plates	
☐	Compare electric and gravitational fields (similarities and differences)	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

The key to electric fields is seeing the deep parallel with gravitational fields: same inverse-square laws, same potential formalism — but with charge replacing mass, and the crucial difference that charge can be positive or negative.

CIE International A Level Physics

Topic 19

Capacitance

Revision Booklet

This booklet covers:

- Capacitors and Capacitance
- Capacitors in Series and Parallel
- Energy Stored in a Capacitor
- Capacitor Discharge
- The Time Constant

45	Core Concepts and Definitions	63
46	Capacitors in Series and Parallel	63
47	Energy Stored in a Capacitor	64
48	Discharging a Capacitor	65
49	Formula Summary Sheet	67
50	Worked Examples	67
51	Practice Exam Questions	68
52	Mark Scheme and Answers	71
53	Revision Checklist	73

45 Core Concepts and Definitions

Capacitance

The **capacitance** of a conductor is the charge stored per unit potential difference across it.

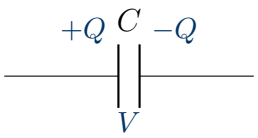
$$C = \frac{Q}{V} \qquad \text{units: F (farads)} \equiv \text{C V}^{-1}$$

- 1 F is a very large unit; practical capacitors are typically μF, nF or pF.
- Capacitance depends on the **geometry** of the conductor and the medium between the plates, not on Q or V individually.
- For an **isolated spherical conductor** of radius R: $C = 4\pi\varepsilon_0 R$.

The Parallel Plate Capacitor

Two conducting plates of area A separated by distance d:

- Charging the capacitor stores charge $+Q$ on one plate and $-Q$ on the other.
- The electric field between the plates is uniform: $E = V/d$.
- Capacitance increases with larger plate area A and smaller separation d.



46 Capacitors in Series and Parallel

Capacitors in Parallel

$$C_{\text{total}} = C_1 + C_2 + C_3 + \cdots$$

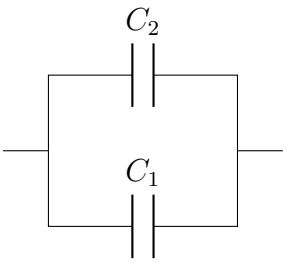
Each capacitor has the **same voltage**; charges add: $Q_{\text{total}} = Q_1 + Q_2 + \cdots$

Capacitors in Series

$$\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots$$

Each capacitor has the **same charge**; voltages add: $V_{\text{total}} = V_1 + V_2 + \cdots$

Circuit diagrams



Parallel: $C_T = C_1 + C_2$

Series: $\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$

Derivation of Series Formula

For capacitors in series, each carries the same charge Q. The total voltage is:

$$V_{\text{total}} = V_1 + V_2 = \frac{Q}{C_1} + \frac{Q}{C_2} = Q\left(\frac{1}{C_1} + \frac{1}{C_2}\right)$$

Since $C_{\text{total}} = Q/V_{\text{total}}$, dividing through by Q gives $\dfrac{1}{C_{\text{total}}} = \dfrac{1}{C_1} + \dfrac{1}{C_2}$.

Common Mistake

Capacitors in series combine like **resistors in parallel** (reciprocal rule), and capacitors in parallel combine like **resistors in series** (direct sum). This is the opposite to resistors — don't mix them up.

47 Energy Stored in a Capacitor

Energy Stored

$$W = \tfrac{1}{2}QV = \tfrac{1}{2}CV^2 = \frac{Q^2}{2C}$$

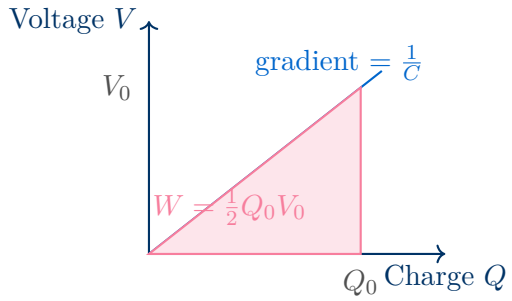
W = energy stored (J)
Q = charge stored (C)
V = potential difference across capacitor (V)
C = capacitance (F)

Energy from the V–Q Graph

The energy stored equals the **area under the V–Q graph** (a straight line through the origin with gradient $1/C$):

$$W = \text{area of triangle} = \tfrac{1}{2} \times Q \times V = \tfrac{1}{2}QV$$

V–Q graph for a capacitor



48 Discharging a Capacitor

Capacitor Discharge Through a Resistor

When a charged capacitor discharges through a resistor R, the charge, voltage and current all decay **exponentially** with time:

$$Q = Q_0\, e^{-t/RC} \qquad V = V_0\, e^{-t/RC} \qquad I = I_0\, e^{-t/RC}$$

where $I_0 = V_0/R = Q_0/RC$ is the initial current.

Time Constant

$$\tau = RC$$

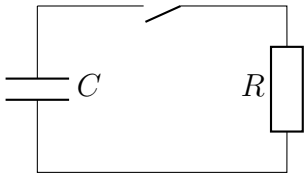
τ = time constant (s)
R = resistance of discharge path (Ω)
C = capacitance (F)

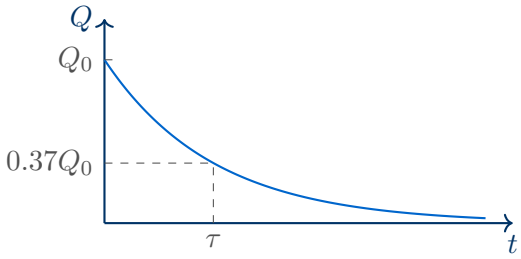
After one time constant ($t = \tau$), the charge/voltage/current has fallen to $e^{-1} \approx$ **37**% of its initial value.

Key Values During Discharge

- $t = \tau$: $Q = 0.37\, Q_0$ (37%)
- $t = 2\tau$: $Q = 0.135\, Q_0$ (13.5%)
- $t = 5\tau$: $Q \approx 0.007\, Q_0$ — capacitor considered fully discharged.
- A larger τ means slower discharge; a smaller τ means faster discharge.

Discharge circuit and Q–t graph





Analysing Discharge Graphs

- A **linear** $\ln Q$ vs t graph confirms exponential decay; gradient $= -1/RC$.
- The time constant τ can be read directly as the time for Q to fall to $0.37\,Q_0$.
- Doubling R or C doubles τ and halves the rate of discharge.

Linearising the discharge: $\ln Q$ against t

Taking logarithms of $Q = Q_0\,e^{-t/RC}$:

$$\ln Q = \ln Q_0 - \frac{1}{RC}\,t$$

This is of the form $y = mx + c$, so a graph of $\ln Q$ against t gives:

- gradient $= -\dfrac{1}{RC} = -\dfrac{1}{\tau}$
- y-intercept $= \ln Q_0$

49 Formula Summary Sheet

Formula	Quantity	Units
$C = Q/V$	Capacitance (definition)	F
$C_T = C_1 + C_2 + \cdots$	Capacitors in parallel	F
$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots$	Capacitors in series	F
$W = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$	Energy stored	J
$\tau = RC$	Time constant	s
$Q = Q_0\, e^{-t/RC}$	Charge during discharge	C
$V = V_0\, e^{-t/RC}$	Voltage during discharge	V
$I = I_0\, e^{-t/RC}$	Current during discharge	A
$\ln Q = \ln Q_0 - \frac{t}{RC}$	Linearised discharge	—

Constants and values: $e^{-1} \approx 0.368$; after 1τ: 37% remains; after 5τ: $< 1\%$ remains

Units check: $[\tau] = [\Omega][\text{F}] = [\text{V A}^{-1}][\text{C V}^{-1}] = [\text{C A}^{-1}] = \text{s}$

50 Worked Examples

Example 1 — Capacitors in Series and Parallel

Question: Three capacitors of 2.0 μF, 3.0 μF and 6.0 μF are connected (a) in parallel and (b) in series. Find the total capacitance in each case.

Solution

(a) Parallel:

$$C_T = 2.0 + 3.0 + 6.0 = \mathbf{11.0}\ \mu\text{F}$$

(b) Series:

$$\frac{1}{C_T} = \frac{1}{2.0} + \frac{1}{3.0} + \frac{1}{6.0} = \frac{3+2+1}{6.0} = \frac{6}{6.0} = 1.0\ \mu\text{F}^{-1}$$

$$C_T = \mathbf{1.0}\ \mu\text{F}$$

Example 2 — Energy Stored

Question: A 470 μF capacitor is charged to 12 V. Calculate (a) the charge stored and (b) the energy stored.

Solution

(a) $Q = CV = 470 \times 10^{-6} \times 12 = \mathbf{5.64 \times 10^{-3}}$ C
(b) $W = \frac{1}{2}CV^2 = \frac{1}{2} \times 470 \times 10^{-6} \times 12^2 = \mathbf{3.38 \times 10^{-2}}$ J

Example 3 — Capacitor Discharge

Question: A 220 μF capacitor charged to 9.0 V discharges through a 47 kΩ resistor. Calculate (a) the time constant and (b) the voltage after 5.0 s.

Solution

(a) $\tau = RC = 47 \times 10^3 \times 220 \times 10^{-6} = \mathbf{10.3}$ s
(b) $V = V_0\, e^{-t/RC} = 9.0 \times e^{-5.0/10.3} = 9.0 \times e^{-0.485} = 9.0 \times 0.616 = \mathbf{5.5}$ V

51 Practice Exam Questions

Section A — Short Answer Questions

Q1. Define capacitance and state its SI unit.

[2 marks]

Q2. A capacitor stores 360 μC of charge when connected to a 12 V supply. Calculate its capacitance.

[2 marks]

Q3. Two capacitors of 4.0 μF and 12 μF are connected in series across a 6.0 V supply.

(a) Calculate the total capacitance. *[2 marks]*

(b) Calculate the total charge stored. *[1 mark]*

(c) Calculate the voltage across the 4.0 μF capacitor. *[2 marks]*

Q4. Explain, with reference to the V–Q graph, why the energy stored in a capacitor is $W = \frac{1}{2}QV$ and not $W = QV$.

[2 marks]

Q5. Define the time constant for a capacitor–resistor discharge circuit and state what fraction of the initial charge remains after two time constants.

[3 marks]

Section B — Longer Structured Questions

Q6. A 100 μF capacitor is charged to 20 V and then discharged through a 25 kΩ resistor.

(a) Calculate the initial charge stored on the capacitor.

[1 mark]

(b) Calculate the initial discharge current.

[2 marks]

(c) Calculate the time constant for the discharge.

[1 mark]

(d) Calculate the charge remaining after 4.0 s.

[2 marks]

(e) The student plots a graph of $\ln(Q/\mathrm{C})$ against t/s. State the gradient and y-intercept of this graph.

[2 marks]

Q7. A 50 μF and a 200 μF capacitor are connected in series and charged from a 15 V supply.

(a) Calculate the combined capacitance.

[2 marks]

(b) Calculate the energy stored in the combination.

[2 marks]

(c) The two capacitors are now reconnected in parallel across the same supply. Calculate the new total energy stored and explain why it differs from part (b).

[3 marks]

52 Mark Scheme and Answers

Q1. Capacitance is the charge stored per unit potential difference [1]; unit: farad (F) or C V^{-1} [1].

Q2. $C = Q/V = 360 \times 10^{-6}/12 = \mathbf{30}$ μF [2].

Q3(a). $\dfrac{1}{C_T} = \dfrac{1}{4.0} + \dfrac{1}{12} = \dfrac{3+1}{12} = \dfrac{4}{12}$ [1]; $C_T = \mathbf{3.0}$ μF [1].

Q3(b). $Q = C_T V = 3.0 \times 10^{-6} \times 6.0 = \mathbf{1.8 \times 10^{-5}}$ C [1].

Q3(c). Same charge on each capacitor in series; $V_1 = Q/C_1 = 1.8 \times 10^{-5}/(4.0 \times 10^{-6}) =$ **4.5** V [2].

Q4. The V–Q graph is a straight line through the origin [1]; the energy is the area under this graph (a triangle), which is $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2}QV$ — not QV because the voltage builds up gradually from 0 to V as charge is stored [1].

Q5. The time constant is the product RC [1]; it is the time taken for the charge (or voltage or current) to fall to $1/e \approx 37\%$ of its initial value [1]; after 2τ: $e^{-2} \approx \mathbf{13.5}\%$ remains [1].

Q6(a). $Q_0 = CV_0 = 100 \times 10^{-6} \times 20 = \mathbf{2.0 \times 10^{-3}}$ C [1].

Q6(b). $I_0 = V_0/R = 20/(25 \times 10^3)$ [1] $= \mathbf{8.0 \times 10^{-4}}$ A [1].

Q6(c). $\tau = RC = 25 \times 10^3 \times 100 \times 10^{-6} = \mathbf{2.5}$ s [1].

Q6(d). $Q = Q_0\, e^{-t/RC} = 2.0{\times}10^{-3}{\times}e^{-4.0/2.5}$ [1] $= 2.0{\times}10^{-3}{\times}e^{-1.6} = 2.0{\times}10^{-3}{\times}0.202 = \mathbf{4.0 \times 10^{-4}}$ C [1].

Q6(e). Gradient $= -1/RC = -1/2.5 = \mathbf{-0.40}$ $\mathbf{s^{-1}}$ [1]; y-intercept $= \ln Q_0 = \ln(2.0 \times 10^{-3}) = \mathbf{-6.2}$ [1].

Q7(a). $\dfrac{1}{C_T} = \dfrac{1}{50} + \dfrac{1}{200} = \dfrac{4+1}{200} = \dfrac{5}{200}$ [1]; $C_T = \mathbf{40}$ μF [1].

Q7(b). $W = \frac{1}{2}C_T V^2 = \frac{1}{2} \times 40 \times 10^{-6} \times 15^2$ [1] $= \mathbf{4.5 \times 10^{-3}}$ J [1].

Q7(c). $C_T = 50 + 200 = 250$ μF; $W = \frac{1}{2} \times 250 \times 10^{-6} \times 15^2$ [1] $= \mathbf{2.8 \times 10^{-2}}$ J [1]; more energy is stored in parallel because the total capacitance is greater — more charge is drawn from the supply [1].

53 Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

	Learning Objective	Confidence (1–3)
☐	Define capacitance; use $C = Q/V$	
☐	Calculate combined capacitance for series and parallel combinations	
☐	Derive the series and parallel formulae from $C = Q/V$	
☐	Use $W = \frac{1}{2}QV = \frac{1}{2}CV^2 = Q^2/2C$ for energy stored	
☐	Explain why energy stored is the area under a V–Q graph	
☐	Describe the exponential decay of Q, V and I during discharge	
☐	Use $x = x_0\, e^{-t/RC}$ for discharge of charge, voltage or current	
☐	Define the time constant $\tau = RC$ and state its physical significance	
☐	Determine τ from a discharge graph (directly or via $\ln Q$ vs t)	
☐	Sketch discharge curves for Q, V and I against time	
☐	Linearise the discharge equation and interpret gradient and intercept	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

The exponential decay of a capacitor is one of the most important mathematical forms in physics. Once you can recognise it, linearise it, and extract τ from a graph, you have a powerful tool that reappears throughout the course.

CIE International A Level Physics

Topic 20

Magnetic Fields

Revision Booklet

This booklet covers:

- Magnetic Fields and Field Lines
- Force on a Current-Carrying Conductor
- Force on a Moving Charge
- Hall Effect and Velocity Selector
- Magnetic Fields due to Currents
- Electromagnetic Induction

54	Magnetic Fields	76
55	Force on a Current-Carrying Conductor	76
56	Force on a Moving Charge	77
57	Magnetic Fields due to Currents	78
58	Electromagnetic Induction	79
59	Formula Summary Sheet	80
60	Worked Examples	80
61	Practice Exam Questions	81
62	Mark Scheme and Answers	83
63	Revision Checklist	85

54 Magnetic Fields

Magnetic Field

A **magnetic field** is a region of space in which a moving charge, or a current-carrying conductor, experiences a force.

- Magnetic fields are produced by **moving charges** (electric currents) or by **permanent magnets**.
- The field is represented by **field lines** (flux lines); the direction is the direction of the force on a north pole.
- Field lines never cross; closer lines indicate a stronger field.
- Crosses ($\times$) represent field into the page; dots ($\bullet$) represent field out of the page.

55 Force on a Current-Carrying Conductor

Force on a Conductor

$$F = BIL\sin\theta$$

F = force on the conductor (N)
B = magnetic flux density (T)
I = current in the conductor (A)
L = length of conductor in the field (m)
θ = angle between the conductor and the field direction

Force is **maximum** when $\theta = 90°$ (conductor perpendicular to field): $F = BIL$.
Force is **zero** when $\theta = 0°$ (conductor parallel to field).

Magnetic Flux Density B

Magnetic flux density is defined as the force per unit current per unit length acting on a wire placed **at right angles** to the field.

$$B = \frac{F}{IL} \qquad \text{units: T (tesla)} \equiv \text{N A}^{-1}\text{m}^{-1}$$

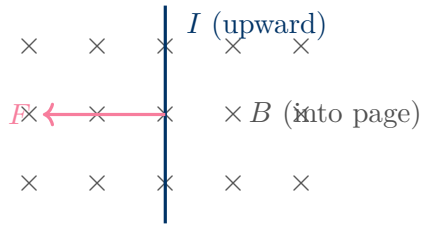
B is a vector quantity. 1 T is a strong field; Earth's field is $\approx 50\ \mu$T.

Fleming's Left-Hand Rule

For a **conventional current** in a magnetic field:

- **First finger**: direction of magnetic field B
- **Second finger**: direction of conventional current I
- **Thumb**: direction of force (motion) F

Remember: the rule gives the force on *positive* charges / conventional current. For electrons, reverse the direction.



56 Force on a Moving Charge

Force on a Moving Charge

$$F = BQv \sin\theta$$

F = magnetic force (N)
B = magnetic flux density (T)
Q = charge (C)
v = speed of the charge (m s^{-1})
θ = angle between velocity and field

For $\theta = 90°$: $F = BQv$, directed perpendicular to both v and B.

Circular Motion in a Magnetic Field

When a charged particle moves **perpendicular** to a uniform magnetic field, the magnetic force is always perpendicular to the velocity, so no work is done and the **speed is constant**. The particle moves in a **circle**:

$$BQv = \frac{mv^2}{r} \quad \Rightarrow \quad r = \frac{mv}{BQ}$$

A larger momentum or smaller B gives a larger radius.

56.1 The Hall Effect

Hall Voltage

When a current-carrying conductor is placed in a magnetic field perpendicular to the current, charge carriers experience a sideways force. They accumulate on one face until the electric force balances the magnetic force, producing the **Hall voltage**:

$$V_H = \frac{BI}{ntq}$$

V_H = Hall voltage (V)
B = magnetic flux density (T)
I = current through the conductor (A)
n = number density of charge carriers (m^{-3})
t = thickness of the conductor in the direction of B (m)
q = charge on each carrier (C)

A **Hall probe** uses this effect to measure B: since $V_H \propto B$ at constant I.

56.2 Velocity Selector

Velocity Selector

A velocity selector uses **crossed** electric and magnetic fields so that only particles with a specific speed pass through undeflected:

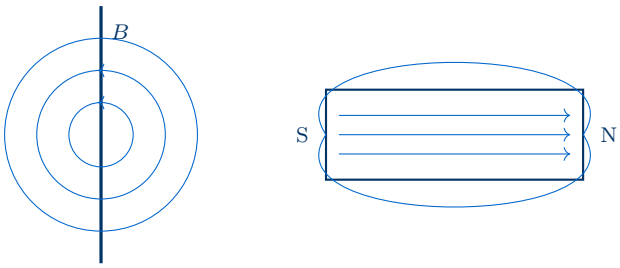
$$qE = BQv \quad \Rightarrow \quad v = \frac{E}{B}$$

- Electric force qE and magnetic force BQv act in **opposite** directions.
- Only particles where these forces balance travel in a straight line.
- Faster particles are deflected one way; slower particles the other.

57 Magnetic Fields due to Currents

Field Patterns

- **Long straight wire**: concentric circles centred on the wire. Direction given by the **right-hand grip rule** — thumb points in direction of current, fingers curl in direction of field.
- **Flat circular coil**: field lines pass through the centre of the coil. Field at the centre is approximately uniform over a small region.
- **Long solenoid**: nearly uniform field inside, similar to a bar magnet outside. A **ferrous (iron) core** greatly increases the field strength.



Straight wire Solenoid

Forces Between Parallel Conductors

- **Same direction currents**: conductors **attract** each other.
- **Opposite direction currents**: conductors **repel** each other.
- Each conductor sits in the magnetic field produced by the other; the force is given by $F = BIL$.

58 Electromagnetic Induction

Magnetic Flux

Magnetic flux Φ is the product of the magnetic flux density and the cross-sectional area perpendicular to the field:

$$\Phi = BA\cos\theta \qquad \text{units: Wb (weber)} \equiv \text{T m}^2$$

When B is perpendicular to the area: $\Phi = BA$.

Flux linkage $N\Phi$ is the flux through a single turn multiplied by the number of turns N of the coil: units Wb-turns.

Faraday's Law and Lenz's Law

$$\mathcal{E} = -\frac{\Delta(N\Phi)}{\Delta t}$$

Faraday's Law: the induced e.m.f. is proportional to the rate of change of flux linkage.

Lenz's Law: the induced e.m.f. (and hence current) acts in a direction that **opposes** the change in flux that caused it (the minus sign above).

$\mathcal{E}$ = induced e.m.f. (V)
N = number of turns
$\Delta\Phi/\Delta t$ = rate of change of flux (Wb s^{-1} $\equiv$ V)

Factors Affecting the Induced E.M.F.

- **Rate of change** of flux: faster change $\Rightarrow$ larger e.m.f.
- **Number of turns** N: more turns $\Rightarrow$ larger e.m.f.
- **Strength of field** B: stronger field $\Rightarrow$ larger flux change.
- **Area** of coil: larger area $\Rightarrow$ more flux.

Lenz's law is a consequence of conservation of energy — the induced current creates a force that opposes the motion causing induction.

Common Mistake

An e.m.f. is induced only when the flux is **changing**. A conductor stationary in a steady field has zero induced e.m.f., even if the field is strong.

59 Formula Summary Sheet

Formula	Quantity	Units
$F = BIL\sin\theta$	Force on current-carrying conductor	N
$F = BQv\sin\theta$	Force on moving charge	N
$r = mv/(BQ)$	Radius of circular orbit in B field	m
$V_H = BI/(ntq)$	Hall voltage	V
$v = E/B$	Velocity selector condition	m s^{-1}
$\Phi = BA\cos\theta$	Magnetic flux	Wb
$\mathcal{E} = -\Delta(N\Phi)/\Delta t$	Faraday's / Lenz's law	V

Units: 1 T = 1 N A^{-1}m^{-1}; 1 Wb = 1 T m^2 = 1 V s
Right-hand grip rule: thumb ∥ current, fingers curl in direction of B field.

60 Worked Examples

Example 1 — Force on a Conductor

Question: A wire of length 0.15 m carries a current of 3.0 A at 60° to a uniform field of 0.25 T. Calculate the force on the wire.

Solution

$$F = BIL\sin\theta = 0.25 \times 3.0 \times 0.15 \times \sin 60^\circ$$
$$F = 0.25 \times 3.0 \times 0.15 \times 0.866 = \mathbf{9.7 \times 10^{-2}}\ \text{N}$$

Example 2 — Circular Motion of a Charged Particle

Question: A proton (mass 1.67×10^{-27} kg, charge 1.6×10^{-19} C) moves at 2.0×10^6 m s^{-1} perpendicular to a field of 0.15 T. Calculate the radius of its circular path.

Solution

$$r = \frac{mv}{BQ} = \frac{1.67 \times 10^{-27} \times 2.0 \times 10^6}{0.15 \times 1.6 \times 10^{-19}} = \frac{3.34 \times 10^{-21}}{2.40 \times 10^{-20}} = \mathbf{0.14}\ \text{m}$$

Example 3 — Induced E.M.F.

Question: A coil of 200 turns and area 50 cm^2 is in a field of 0.30 T perpendicular to the plane of the coil. The field drops to zero in 0.040 s. Calculate the induced e.m.f.

Solution

$$\Delta\Phi = B \times A = 0.30 \times 50 \times 10^{-4} = 1.5 \times 10^{-3} \text{ Wb}$$
$$\mathcal{E} = N\frac{\Delta\Phi}{\Delta t} = 200 \times \frac{1.5 \times 10^{-3}}{0.040} = \mathbf{7.5} \text{ V}$$

61 Practice Exam Questions

Section A — Short Answer Questions

Q1. Define magnetic flux density and state its SI unit.

[2 marks]

Q2. A wire of length 8.0 cm is placed perpendicular to a field of flux density 0.40 T and carries a current of 2.5 A. Calculate the force on the wire.

[2 marks]

Q3. State Faraday's Law and Lenz's Law of electromagnetic induction.

[3 marks]

Q4. Explain why the speed of a charged particle moving perpendicular to a uniform magnetic field remains constant.

[2 marks]

Q5. A Hall probe gives a voltage of 3.2 mV when a current of 50 mA passes through a slice of thickness 2.0 mm in a field B. Given $n = 8.5 \times 10^{28}$ m^{-3} and $q = 1.6 \times 10^{-19}$ C, calculate B.

[3 marks]

Section B — Longer Structured Questions

Q6. An electron (mass 9.11×10^{-31} kg, charge 1.6×10^{-19} C) enters a uniform magnetic field of 2.0×10^{-3} T perpendicular to the field with speed 5.0×10^{6} m s^{-1}.

(a) Calculate the radius of the circular path followed by the electron.

[2 marks]

(b) The field strength is doubled. State and explain the effect on the radius.

[2 marks]

(c) An electric field is now applied perpendicular to B so that the electron travels in a straight line. Calculate the electric field strength required.

[2 marks]

Q7. A rectangular coil of 80 turns and dimensions 4.0 cm × 6.0 cm is placed with its plane perpendicular to a uniform field of 0.50 T.

(a) Calculate the flux linkage through the coil.

[2 marks]

(b) The coil is rotated so that its plane becomes parallel to the field in 0.030 s. Calculate the mean induced e.m.f.

[2 marks]

(c) State and explain the direction of the induced current using Lenz's law.

[2 marks]

62 Mark Scheme and Answers

Q1. Magnetic flux density is the force per unit current per unit length on a wire placed at right angles to the field [1]; unit: tesla (T) or N A^{-1} m^{-1} [1].

Q2. $F = BIL\sin 90° = 0.40 \times 2.5 \times 0.080 =$ **0.080** N [2].

Q3. Faraday's Law: the induced e.m.f. is proportional to the rate of change of flux linkage [1]; $\mathcal{E} = -\Delta(N\Phi)/\Delta t$ [1]. Lenz's Law: the induced e.m.f. acts in a direction to oppose the change in flux that caused it [1].

Q4. The magnetic force is always perpendicular to the velocity [1]; a perpendicular force does no work, so kinetic energy and hence speed remain unchanged [1].

Q5. $B = V_H ntq/I = (3.2 \times 10^{-3} \times 8.5 \times 10^{28} \times 2.0 \times 10^{-3} \times 1.6 \times 10^{-19})/(50 \times 10^{-3})$ [2] = **0.174** T [1].

Q6(a). $r = mv/(BQ) = (9.11 \times 10^{-31} \times 5.0 \times 10^{6})/(2.0 \times 10^{-3} \times 1.6 \times 10^{-19})$ [1] $= \mathbf{1.4 \times 10^{-2}}$ m [1].

Q6(b). Radius halves [1]; $r = mv/BQ$, so $r \propto 1/B$; doubling B halves r [1].

Q6(c). For straight-line motion: $qE = BQv$; $E = Bv = 2.0 \times 10^{-3} \times 5.0 \times 10^{6} = \mathbf{1.0 \times 10^{4}}$ V m$^{\mathbf{-1}}$ [2].

Q7(a). $\Phi = BA = 0.50 \times (0.040 \times 0.060) = 1.2 \times 10^{-3}$ Wb [1]; flux linkage $= N\Phi = 80 \times 1.2 \times 10^{-3} = \mathbf{9.6 \times 10^{-2}}$ Wb-turns [1].

Q7(b). $\Delta(N\Phi) = 9.6 \times 10^{-2}$ Wb-turns (falls to zero) [1]; $\mathcal{E} = 9.6 \times 10^{-2}/0.030 = \mathbf{3.2}$ V [1].

Q7(c). By Lenz's law the induced current opposes the decrease in flux [1]; the current flows in the direction that would create a field to oppose the rotation / maintain the flux through the coil [1].

63 Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

	Learning Objective	Confidence (1–3)
☐	Define magnetic flux density; use $F = BIL\sin\theta$	
☐	Apply Fleming's left-hand rule to find force directions	
☐	Use $F = BQv\sin\theta$ for force on a moving charge	
☐	Derive and use $r = mv/(BQ)$ for circular orbit radius	
☐	Explain and use the Hall effect; use $V_H = BI/(ntq)$	
☐	Explain the velocity selector condition $v = E/B$	
☐	Sketch field patterns for straight wire, circular coil and solenoid	
☐	Apply the right-hand grip rule for field direction around a wire	
☐	Explain forces between parallel current-carrying conductors	
☐	Define magnetic flux $\Phi = BA\cos\theta$ and flux linkage $N\Phi$	
☐	State and apply Faraday's and Lenz's laws	
☐	Identify factors that affect the magnitude of induced e.m.f.	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

Faraday's law — one equation — explains the generator, the transformer, and the electric motor. Master flux linkage and you understand how almost all electrical power is generated.

CIE International A Level Physics

Topic 21

Alternating Currents

Revision Booklet

This booklet covers:

- Characteristics of Alternating Currents
- R.M.S. Values and Power
- Half-Wave Rectification
- Full-Wave (Bridge) Rectification
- Smoothing with a Capacitor

64	Characteristics of Alternating Currents	88
65	R.M.S. Values and Power	88
66	Rectification	89
67	Smoothing with a Capacitor	90
68	Formula Summary Sheet	91
69	Worked Examples	91
70	Practice Exam Questions	92
71	Mark Scheme and Answers	95
72	Revision Checklist	96

64 Characteristics of Alternating Currents

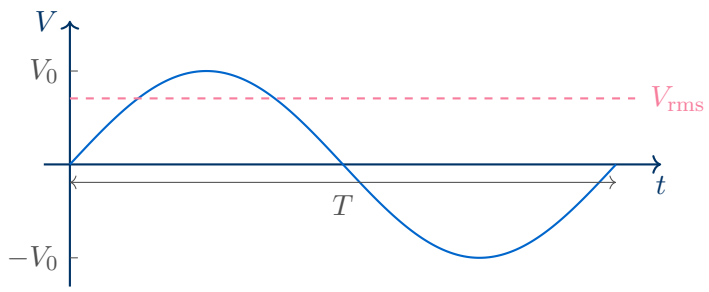
Alternating Current

An **alternating current** (a.c.) reverses direction periodically. For a sinusoidal a.c.:

$$I = I_0 \sin \omega t \qquad V = V_0 \sin \omega t$$

- I_0, V_0: **peak** (amplitude) values.
- $\omega = 2\pi f = 2\pi/T$: angular frequency (rad s^{-1}).
- T: period (s); f: frequency (Hz).
- UK mains supply: $f = 50$ Hz, $T = 20$ ms, $V_{\text{rms}} = 230$ V.

Sinusoidal a.c. voltage waveform



65 R.M.S. Values and Power

Root-Mean-Square (R.M.S.) Value

The **r.m.s. value** of an alternating current is defined as the value of steady direct current that would dissipate the **same power** in a purely resistive load.

R.M.S. Formulae

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \qquad V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$$

These apply to **sinusoidal** waveforms only. For other waveforms the factor $1/\sqrt{2}$ changes.

Mean Power in a Resistive Load

$$P_{\text{mean}} = \tfrac{1}{2}P_{\text{max}} = \tfrac{1}{2}I_0^2 R = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R}$$

The mean power is **half** the peak power for a sinusoidal waveform.

Why Use R.M.S.?

- The mean value of a sinusoidal current is **zero** — useless for power calculations.
- R.M.S. values allow direct use of d.c. power formulae ($P = IV$, $P = I^2R$, $P = V^2/R$).
- Meters and ratings (e.g. 230 V mains) always quote r.m.s. values.

Common Mistake

Do not use peak values in power calculations. Always convert to r.m.s. first: $V_{\text{rms}} = V_0/\sqrt{2}$. The peak mains voltage is $230\sqrt{2} \approx 325$ V, not 230 V.

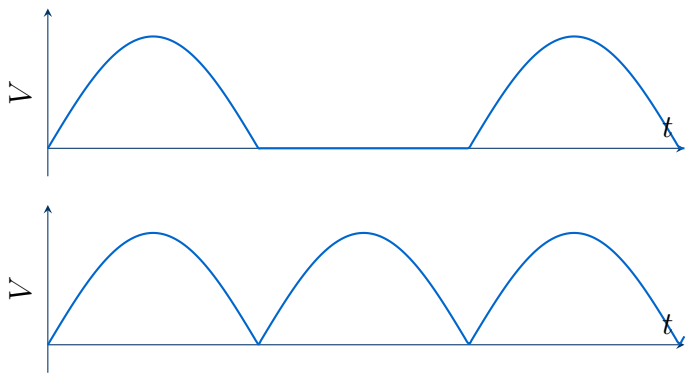
66 Rectification

Rectification

Rectification converts alternating current into direct current flowing in one direction only.

- **Half-wave rectification**: a single diode passes only the positive (or negative) half-cycles; output is a series of pulses with gaps.
- **Full-wave rectification**: a bridge rectifier (four diodes) inverts the negative half-cycles, producing a continuous pulsating d.c. output with no gaps.

Comparison of rectified outputs

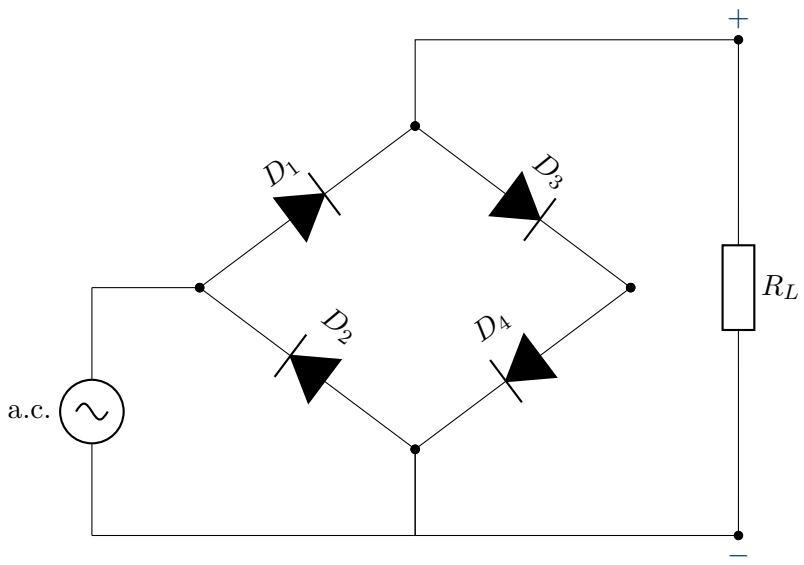


66.1 Bridge Rectifier

Bridge Rectifier

A bridge rectifier uses **four diodes** arranged in a diamond so that:

- During the **positive half-cycle**: current flows through diodes D1 and D4, through the load in the positive direction.
- During the **negative half-cycle**: current flows through diodes D2 and D3, but still through the load in the **same** direction.
- Both half-cycles contribute to the output — no wasted half-cycles.



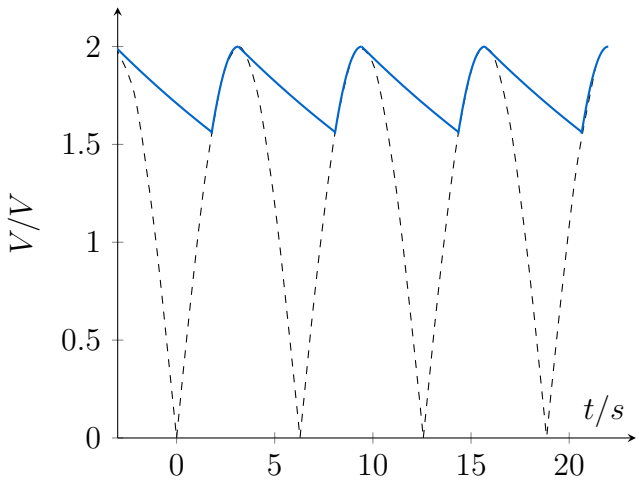
67 Smoothing with a Capacitor

Smoothing

A capacitor connected in **parallel with the load** reduces the ripple on a rectified output:

- The capacitor **charges up** rapidly to the peak voltage during each pulse.
- It **discharges slowly** through the load R_L between peaks, maintaining a more constant voltage.
- The output has a small residual **ripple voltage** rather than falling to zero between pulses.
- **Larger** C or **larger** R_L: longer time constant $\tau = CR_L$, less ripple.
- **Smaller** C or **smaller** R_L: faster discharge, larger ripple.

Effect of smoothing capacitor on full-wave rectified output



Common Mistake

In exam questions, always link ripple size explicitly to the time constant $\tau = CR_L$ — just saying "bigger capacitor" without explanation will not earn full marks.

68 Formula Summary Sheet

Formula	Quantity	Units
$I = I_0 \sin \omega t$	Sinusoidal alternating current	A
$V = V_0 \sin \omega t$	Sinusoidal alternating voltage	V
$\omega = 2\pi f = 2\pi/T$	Angular frequency	rad s^{-1}
$I_{\text{rms}} = I_0/\sqrt{2}$	R.M.S. current	A
$V_{\text{rms}} = V_0/\sqrt{2}$	R.M.S. voltage	V
$P_{\text{mean}} = \frac{1}{2} I_0^2 R$	Mean power (peak values)	W
$P_{\text{mean}} = I_{\text{rms}}^2 R = V_{\text{rms}}^2/R$	Mean power (r.m.s. values)	W

UK mains: $V_{\text{rms}} = 230$ V; $f = 50$ Hz; $V_0 = 230\sqrt{2} \approx 325$ V
Note: $1/\sqrt{2} \approx 0.707$; $P_{\text{mean}} = \frac{1}{2} P_{\text{max}}$ for sinusoidal waveform only.

69 Worked Examples

Example 1 — R.M.S. and Peak Values

Question: The mains supply has $V_{\text{rms}} = 230$ V at 50 Hz. Find (a) the peak voltage, (b) the angular frequency and (c) the mean power in a 1.2 kΩ resistor.

Solution

(a) $V_0 = V_{\text{rms}}\sqrt{2} = 230\sqrt{2} = \mathbf{325}$ V

(b) $\omega = 2\pi f = 2\pi \times 50 = \mathbf{314}$ rad s^{-1}

(c) $P = V_{\text{rms}}^2/R = 230^2/(1.2 \times 10^3) = 52900/1200 = \mathbf{44}$ W

Example 2 — Peak Power and Mean Power

Question: An a.c. supply has peak voltage $V_0 = 12$ V and is connected to a 60 Ω resistor. Calculate (a) the peak power and (b) the mean power dissipated.

Solution

(a) $P_{\text{max}} = V_0^2/R = 144/60 = \mathbf{2.4}$ W

(b) $P_{\text{mean}} = \frac{1}{2}P_{\text{max}} = \mathbf{1.2}$ W

Or equivalently: $V_{\text{rms}} = 12/\sqrt{2}$; $P = V_{\text{rms}}^2/R = 72/60 = 1.2$ W

Example 3 — Smoothing

Question: A full-wave rectifier feeds a 470 μF smoothing capacitor in parallel with a 2.2 kΩ load. Calculate the time constant and comment on the degree of smoothing for a 50 Hz supply.

Solution

$\tau = CR = 470 \times 10^{-6} \times 2200 = \mathbf{1.03}$ s

The period of the full-wave rectified signal is $T/2 = 1/(2 \times 50) = 10$ ms.

Since $\tau = 1.03$ s $\gg$ 10 ms, the capacitor discharges very little between peaks — the output is well smoothed with very small ripple.

70 Practice Exam Questions

Section A — Short Answer Questions

Q1. Explain what is meant by the r.m.s. value of an alternating current and state why it is more useful than the peak value.

[3 marks]

Q2. An a.c. supply has peak voltage 340 V. Calculate (a) the r.m.s. voltage and (b) the mean power delivered to a 680 Ω resistor.

[3 marks]

Q3. Distinguish between half-wave and full-wave rectification. State the number of diodes required for each.

[3 marks]

Q4. Explain how a capacitor connected in parallel with a load resistor smooths a rectified output. State the effect of increasing the capacitance.

[3 marks]

Section B — Longer Structured Questions

Q5. The alternating voltage from a supply is given by $V = 170\sin(100\pi t)$, where V is in volts and t is in seconds.

(a) State the peak voltage and the frequency of the supply.

[2 marks]

(b) Calculate the r.m.s. voltage.

[1 mark]

(c) The supply is connected to a 500 Ω resistor. Calculate the mean power dissipated.

[2 marks]

(d) The supply is now passed through a bridge rectifier and a smoothing capacitor of 1000 μF is connected in parallel with the 500 Ω load. Calculate the time constant and comment on the effectiveness of smoothing.

[3 marks]

Q6. The graph below represents the output of a full-wave rectifier before smoothing. The peak voltage is 12 V and the supply frequency is 50 Hz.

(a) State the frequency of the rectified output.

[1 mark]

(b) On the same axes, sketch the output after connecting a large smoothing capacitor in parallel with the load. Indicate the approximate ripple voltage.

[2 marks]

(c) Explain what happens to the smoothing if the load resistance is reduced.

[2 marks]

71 Mark Scheme and Answers

Q1. The r.m.s. value is the equivalent steady d.c. that dissipates the same power in a resistive load [1]; it is more useful because power formulae ($P = I^2R$, $P = V^2/R$) can be applied directly [1]; the mean of a sinusoidal current is zero, so it gives no information about power [1].

Q2(a). $V_{\text{rms}} = V_0/\sqrt{2} = 340/\sqrt{2} = \mathbf{240}$ V [1]. **Q2(b).** $P = V_{\text{rms}}^2/R = 240^2/680 = 57600/680 = \mathbf{85}$ W [2].

Q3. Half-wave: uses **1 diode**; only one half of each cycle is passed; output has gaps of zero voltage [1]. Full-wave: uses **4 diodes** (bridge rectifier); both half-cycles are used; output is always positive with no gaps [2].

Q4. The capacitor charges to the peak voltage during each pulse [1]; between pulses it discharges slowly through the load, maintaining a more constant output voltage [1]; increasing C increases the time constant $\tau = CR$, so the capacitor discharges less between pulses and the ripple is smaller [1].

Q5(a). Peak voltage $V_0 = \mathbf{170}$ V [1]; $\omega = 100\pi$, so $f = \omega/2\pi = 100\pi/2\pi = \mathbf{50}$ Hz [1].

Q5(b). $V_{\text{rms}} = 170/\sqrt{2} = \mathbf{120}$ V [1].

Q5(c). $P = V_{\text{rms}}^2/R = (120)^2/500$ [1] $= 14400/500 = \mathbf{28.8}$ W [1].

Q5(d). $\tau = CR = 1000 \times 10^{-6} \times 500 = \mathbf{0.50}$ s [1]; period of full-wave output $= 1/(2 \times 50) = 10$ ms [1]; $\tau \gg T/2$ so capacitor barely discharges between peaks — very effective smoothing with tiny ripple [1].

Q6(a). The full-wave rectified output has frequency $2 \times 50 = \mathbf{100}$ Hz [1].

Q6(b). Sketch: smoothed output just below 12 V with small sawtooth ripple; ripple voltage is the small oscillation between the capacitor charge and discharge levels [2].

Q6(c). Reducing load resistance increases the discharge current [1]; the capacitor discharges faster between peaks ($\tau = CR$ decreases), so the ripple voltage increases and smoothing is less effective [1].

72 Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

	Learning Objective	Confidence (1–3)
☐	Define period, frequency and peak value for an a.c. waveform	
☐	Use $x = x_0 \sin \omega t$ for sinusoidal current or voltage	
☐	Define r.m.s. value and explain why it is useful	
☐	Use $I_{\text{rms}} = I_0/\sqrt{2}$ and $V_{\text{rms}} = V_0/\sqrt{2}$	
☐	Calculate mean power using r.m.s. values: $P = I_{\text{rms}}^2 R = V_{\text{rms}}^2/R$	
☐	State that mean power is half peak power for sinusoidal a.c.	
☐	Distinguish half-wave and full-wave rectification graphically	
☐	Explain the action of a single diode for half-wave rectification	
☐	Explain the action of a bridge rectifier (four diodes)	
☐	Explain smoothing by a capacitor in terms of charge/discharge	
☐	Analyse the effect of C and R_L on the time constant and ripple	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

R.M.S. values are one of physics's most elegant ideas: a way to make a continuously varying quantity equivalent to a steady one. Once you see that $P_{\text{mean}} = \frac{1}{2}P_{\text{max}}$ comes directly from $\langle \sin^2 \rangle = \frac{1}{2}$, the whole topic falls into place.

CIE International A Level Physics

Topic 22

Quantum Physics

Revision Booklet

This booklet covers:

- Energy and Momentum of a Photon
- The Photoelectric Effect
- Wave-Particle Duality
- Energy Levels and Line Spectra

73	**Energy and Momentum of a Photon**	**99**
74	**The Photoelectric Effect**	**99**
75	**Wave-Particle Duality**	**99**
76	**Energy Levels and Line Spectra**	**99**

73 Energy and Momentum of a Photon

74 The Photoelectric Effect

75 Wave-Particle Duality

76 Energy Levels and Line Spectra

Good luck with your revision!

Remember: understanding *why* formulas work is more powerful than memorising them.

CIE International A Level Physics

Topic 23

Nuclear Physics

Revision Booklet

This booklet covers:

- Mass Defect and Binding Energy
- Nuclear Fission and Fusion
- Radioactive Decay

77 Mass Defect and Binding Energy **102**
78 Nuclear Fission and Fusion **102**
79 Radioactive Decay **102**

77 Mass Defect and Binding Energy

78 Nuclear Fission and Fusion

79 Radioactive Decay

Good luck with your revision!

Remember: understanding *why* formulas work is more powerful than memorising them.

CIE International A Level Physics

Topic 24

Medical Physics

Revision Booklet

This booklet covers:

- Production and Use of Ultrasound
- Production and Use of X-rays
- PET Scanning

80 Production and Use of Ultrasound 105
81 Production and Use of X-rays 105
82 PET Scanning 105

80 Production and Use of Ultrasound

81 Production and Use of X-rays

82 PET Scanning

Good luck with your revision!

Remember: understanding *why* formulas work is more powerful than memorising them.

CIE International A Level Physics

Topic 25

Astronomy and Cosmology

Revision Booklet

This booklet covers:

- Standard Candles
- Stellar Radii
- Hubble's Law and the Big Bang

83 Standard Candles 108
84 Stellar Radii 108
85 Hubble's Law and the Big Bang 108

83 Standard Candles

84 Stellar Radii

85 Hubble's Law and the Big Bang

Good luck with your revision!

Remember: understanding *why* formulas work is more powerful than memorising them.